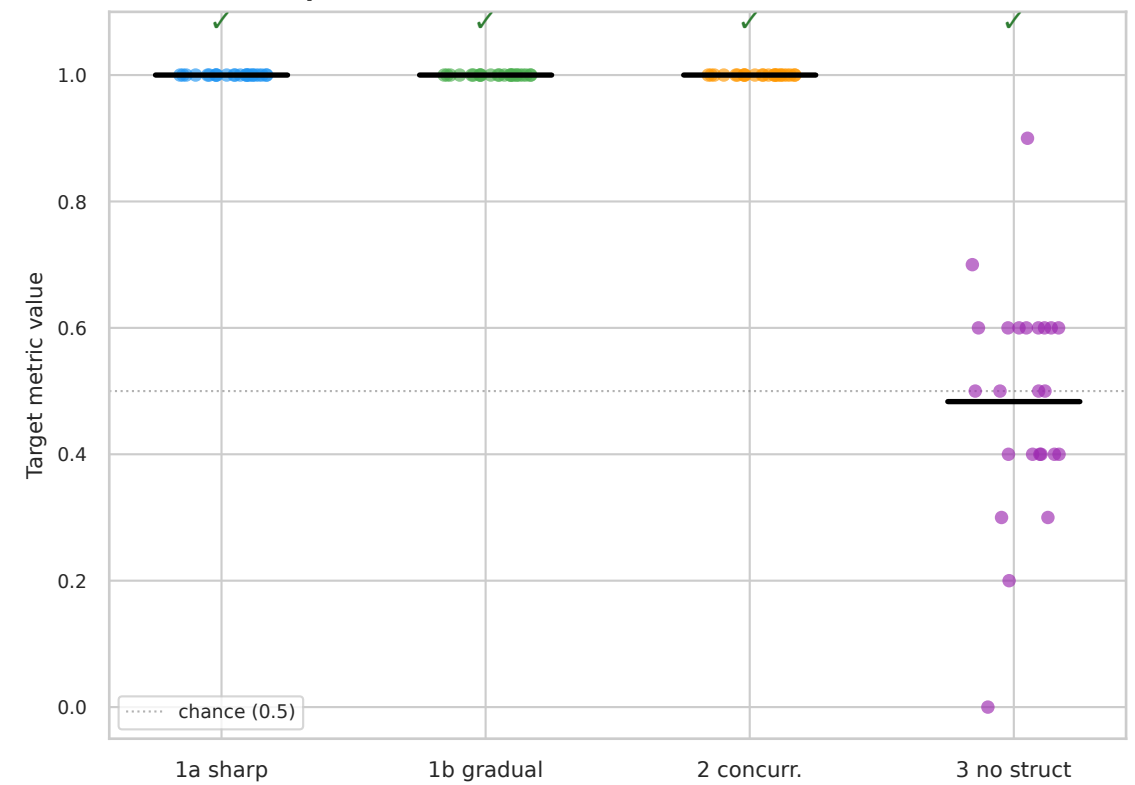
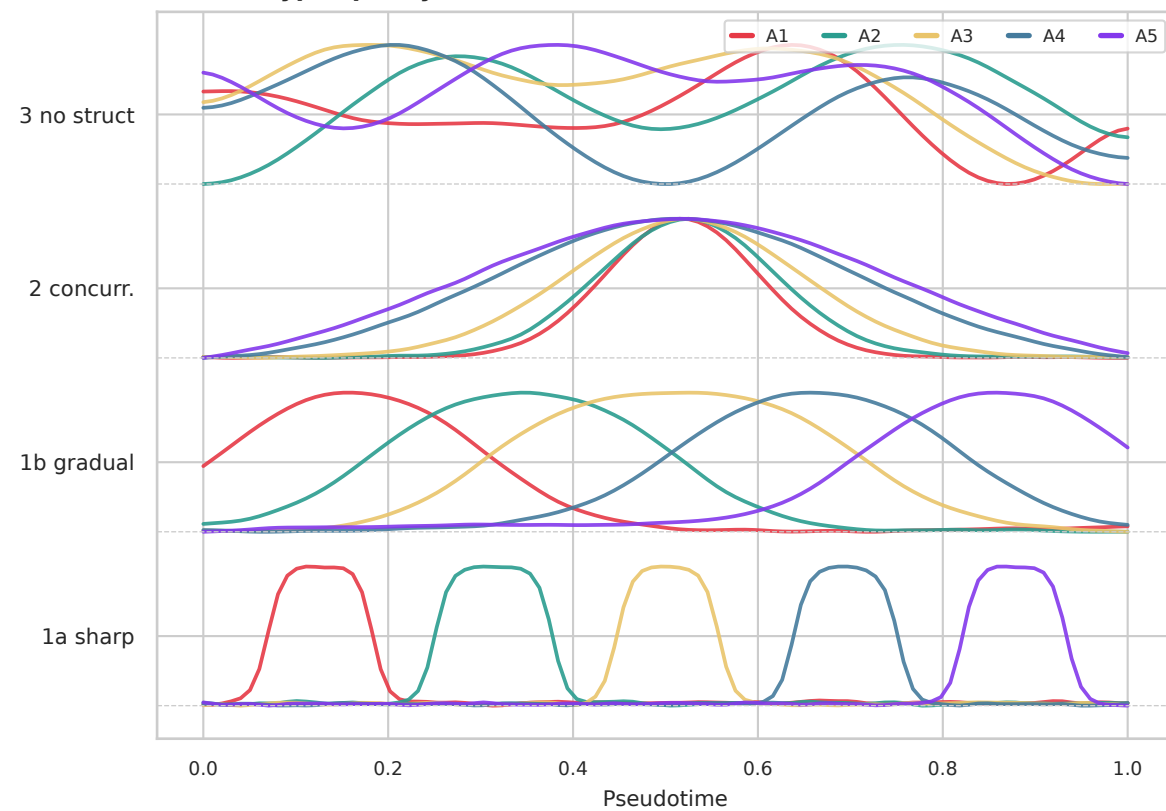


Figure 2 | Validation of the shape-agnostic peak-timing coordination metric

a | Synthetic activation profiles
(5 archetypes per system, one seed)

Synthetic positive-control benchmark — metric validation only
(24 seeds, 200 null replicates; decomposition recovery in Figs 3-4)

b | Peak-timing target metric
across 24 independent seeds



c | Real vs null: target metric per system
(♦ = real mean across seeds; bars = null mean $\pm$ SD)

d | Empirical p-values: real vs null
(target metric per system and null model)

$\alpha = 0.01$ (min possible $p = 0.00498$)

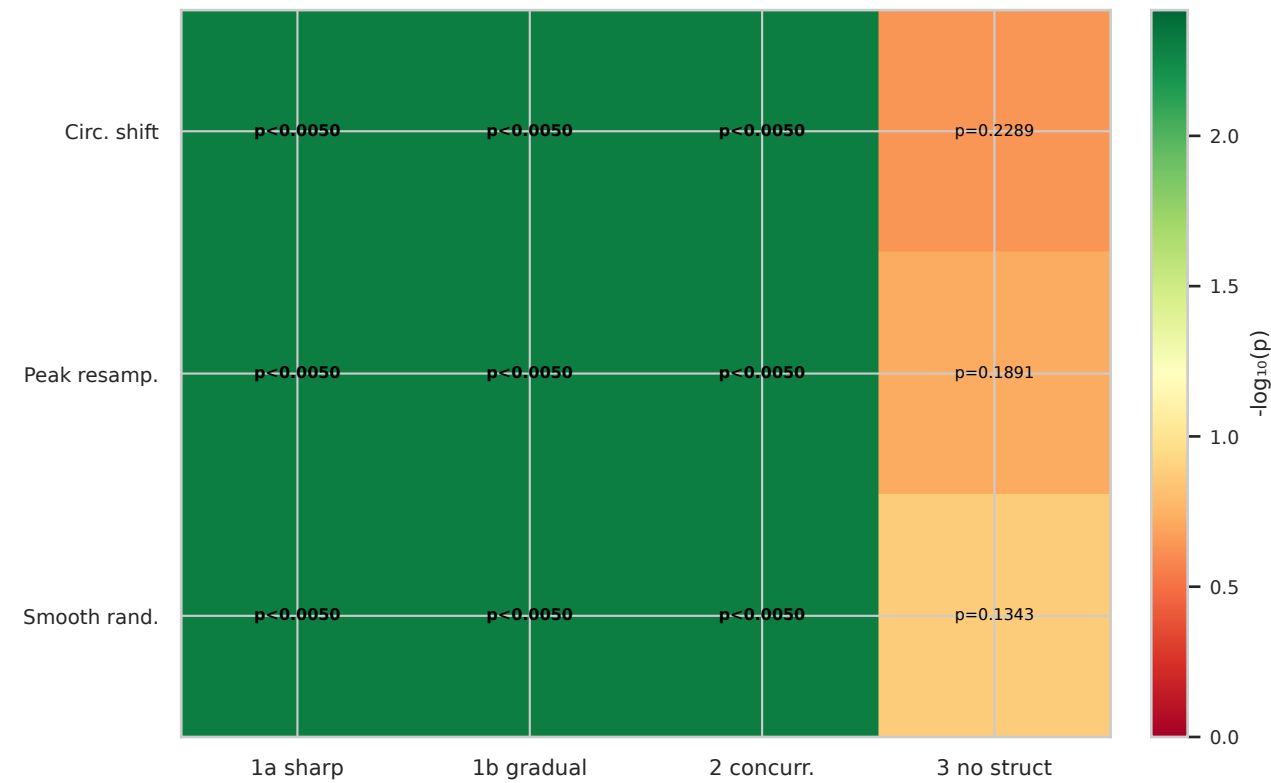
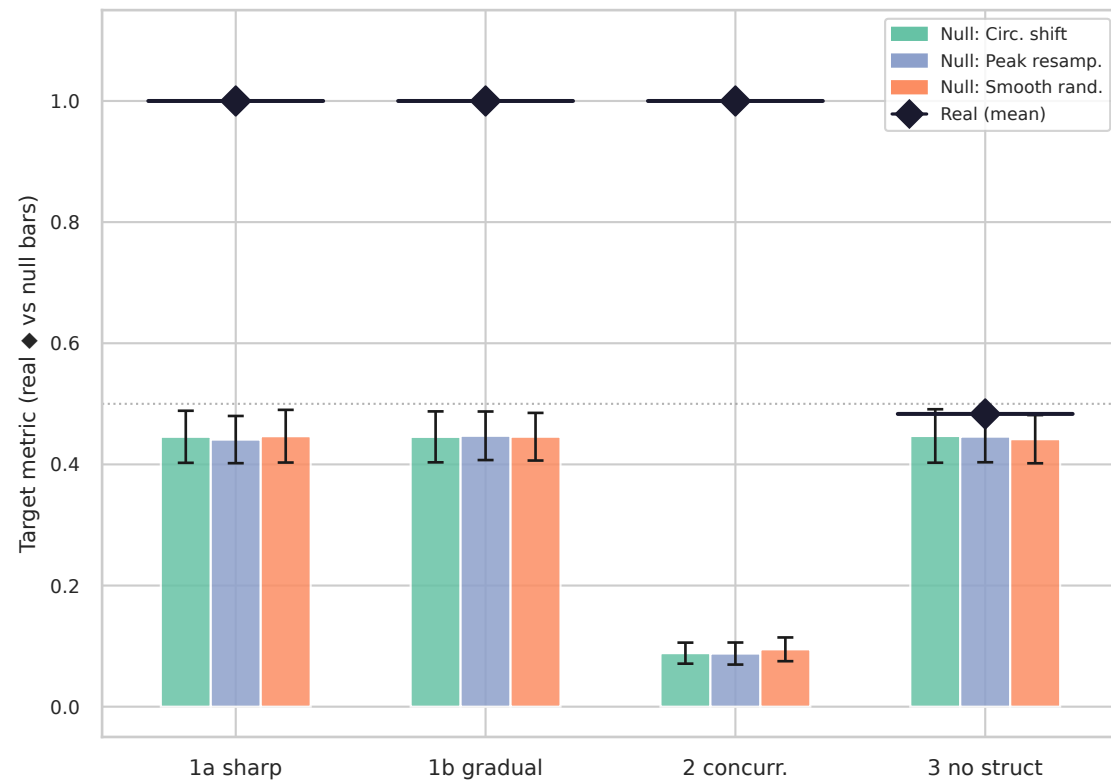
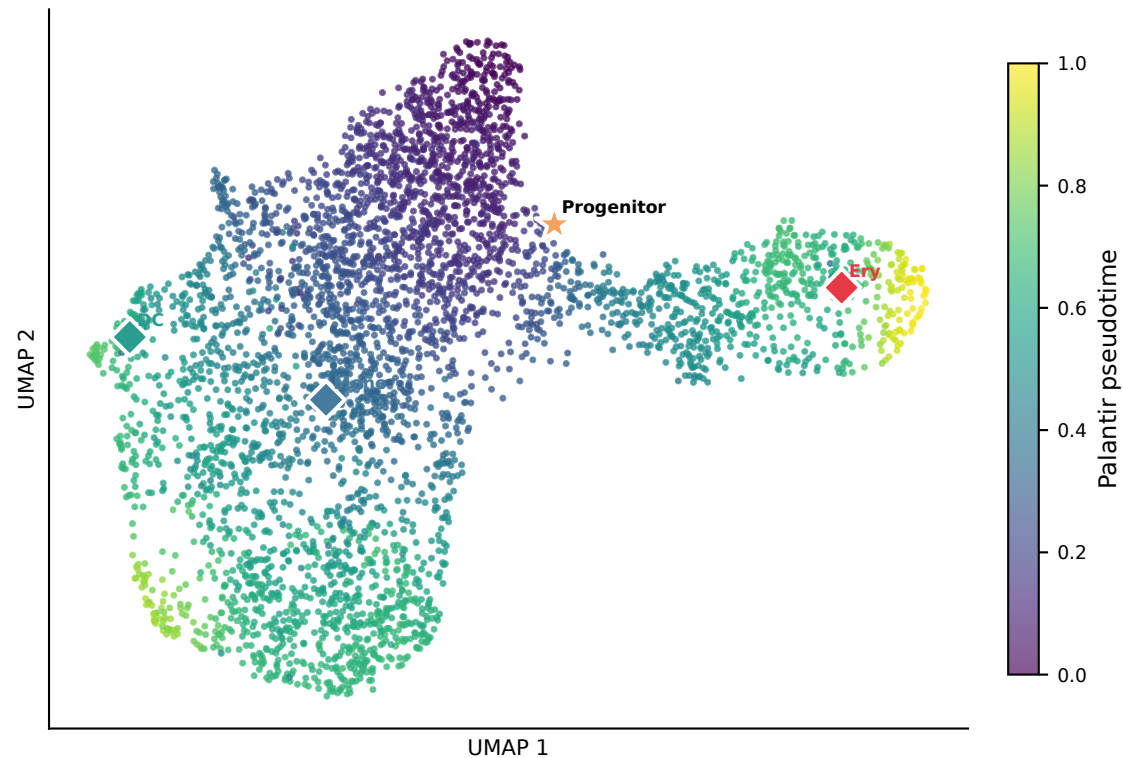


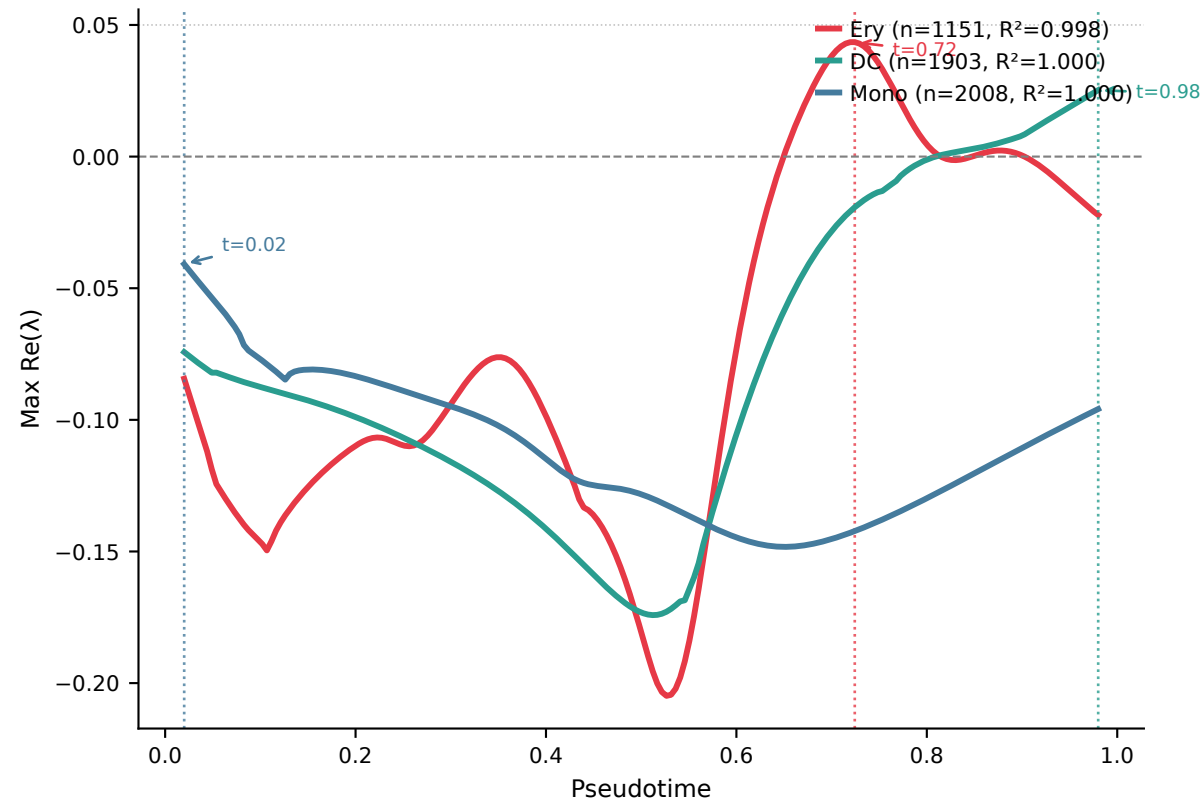
Figure 3 (FA) | Bone Marrow Hematopoiesis — Branch-Separated scJDO

FactorAnalysis (30-D) + UMAP + Palantir pseudotime + biology-informed drift fields | Ery $R^2=0.998$ DC $R^2=1.000$ Mono $R^2=1.000$

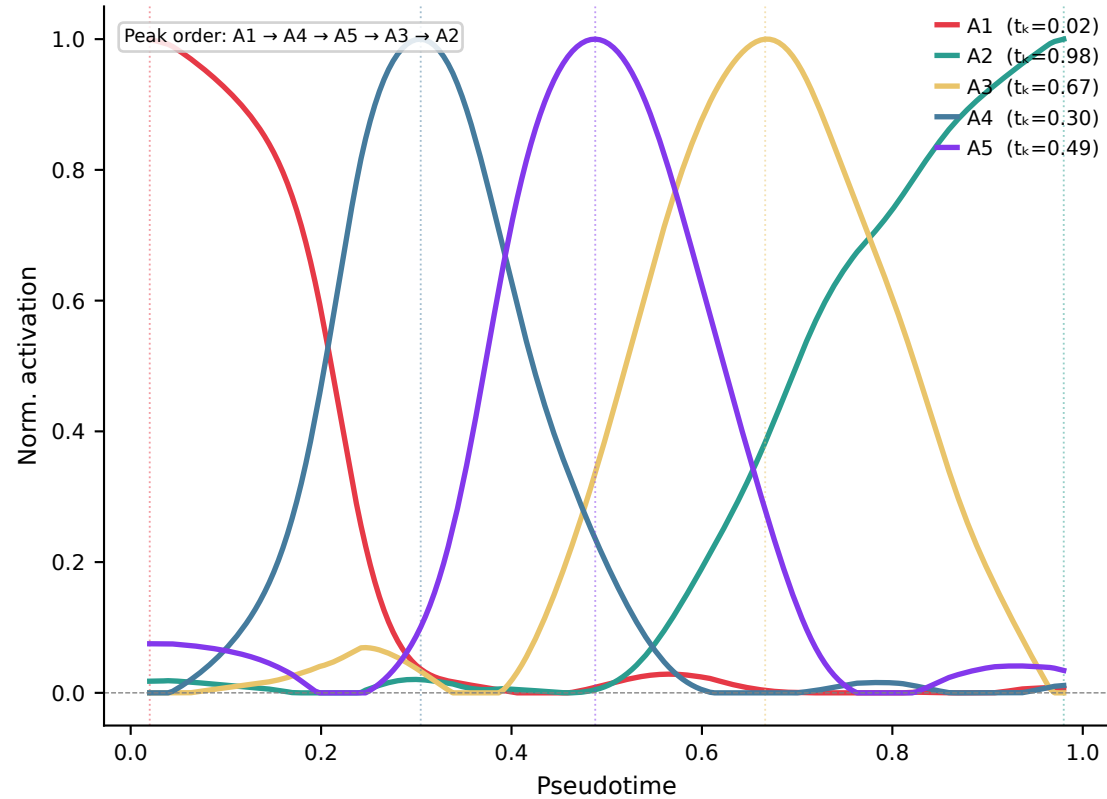
a | Palantir trajectory — Bone Marrow (4,142 cells)



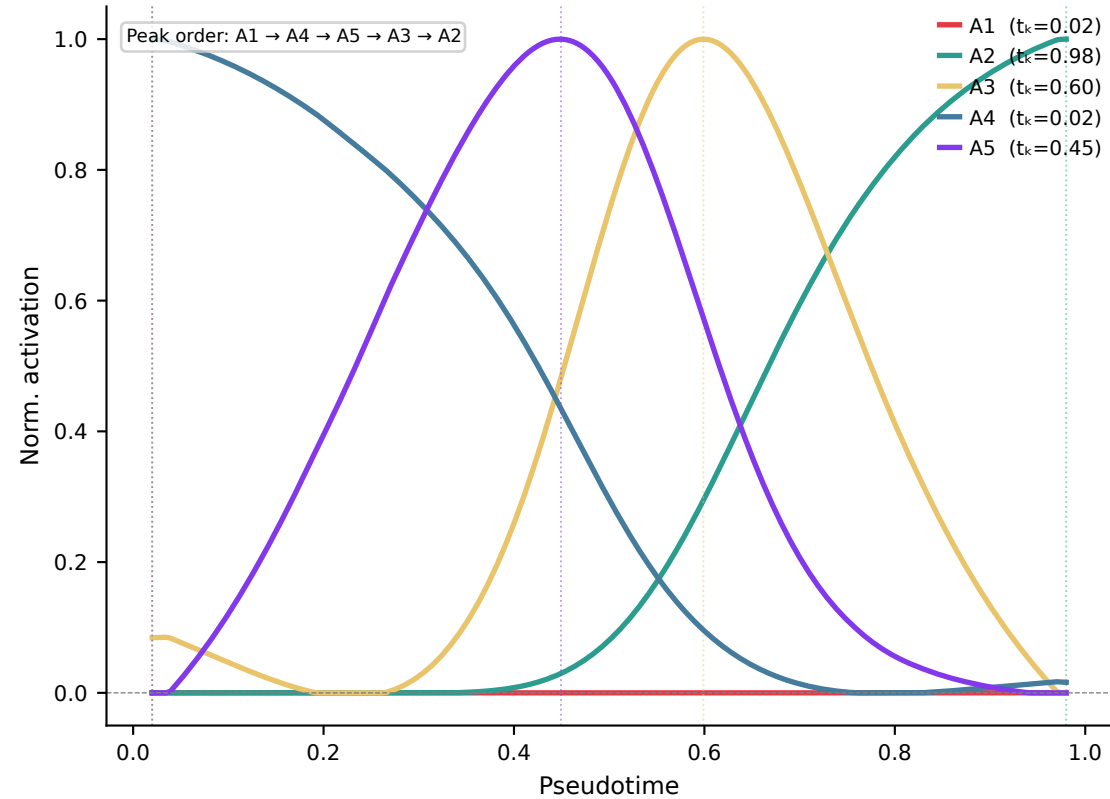
b | Local sensitivity — Ery vs DC vs Mono



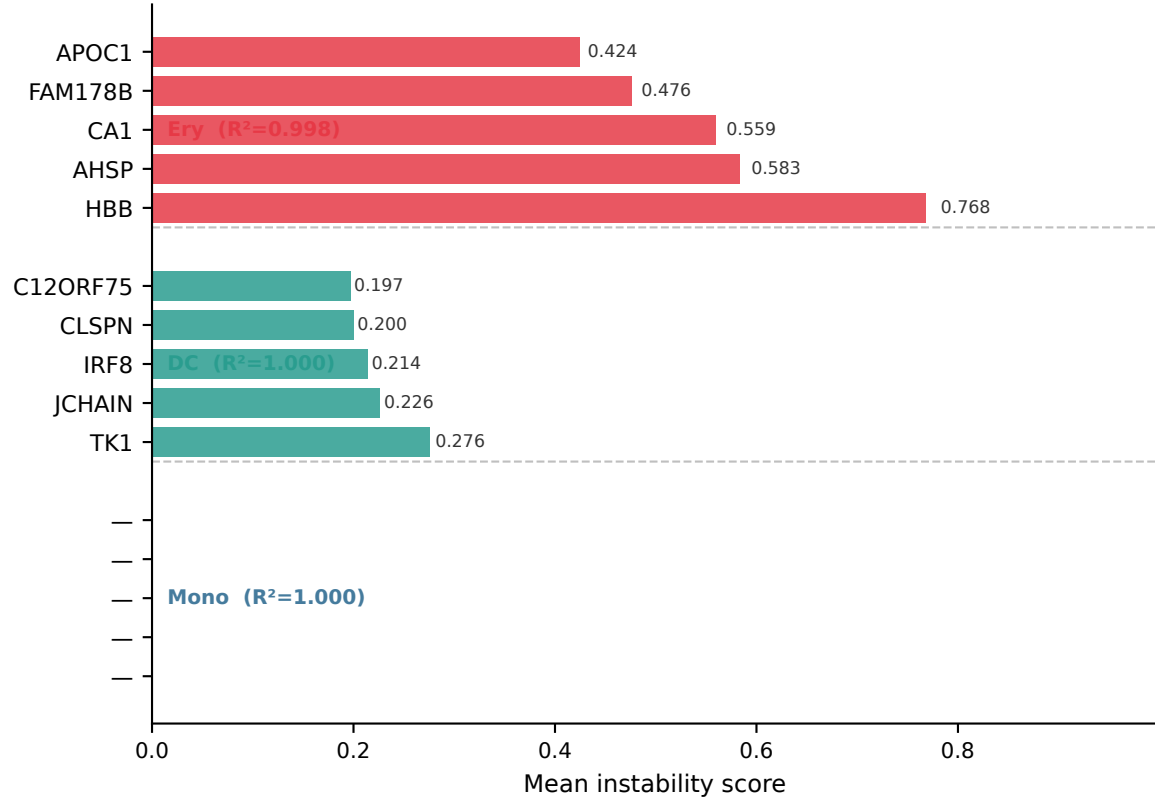
c | Ery archetypes (K=5, $R^2=0.998$, n=1151)



d | DC archetypes (K=5, $R^2=1.000$, n=1903)



e | Top instability genes — Ery / DC / Mono



f | Mono archetypes (K=5, $R^2=1.000$, n=2008)

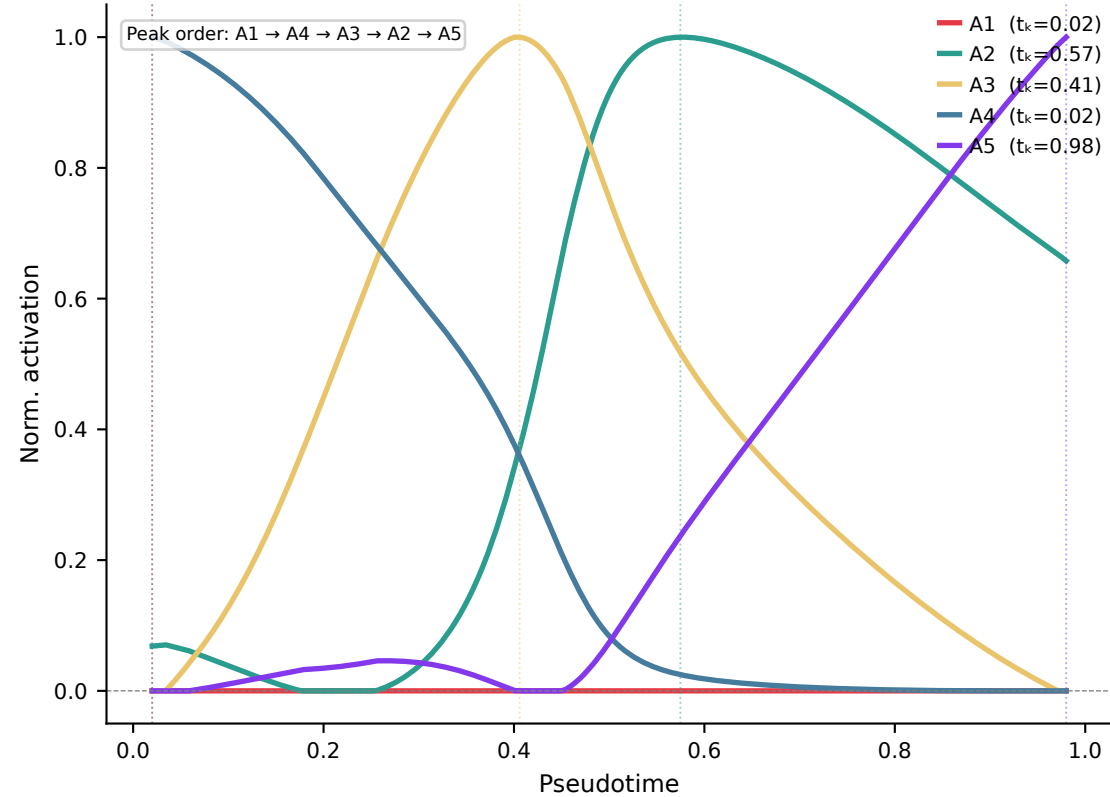
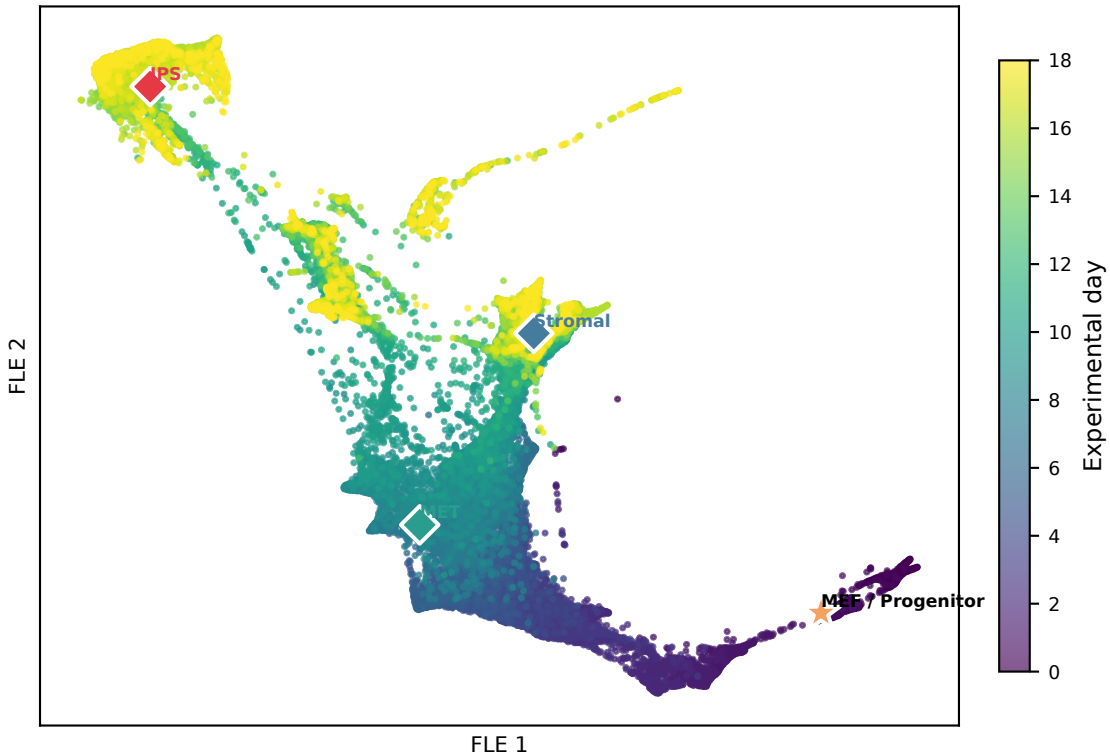


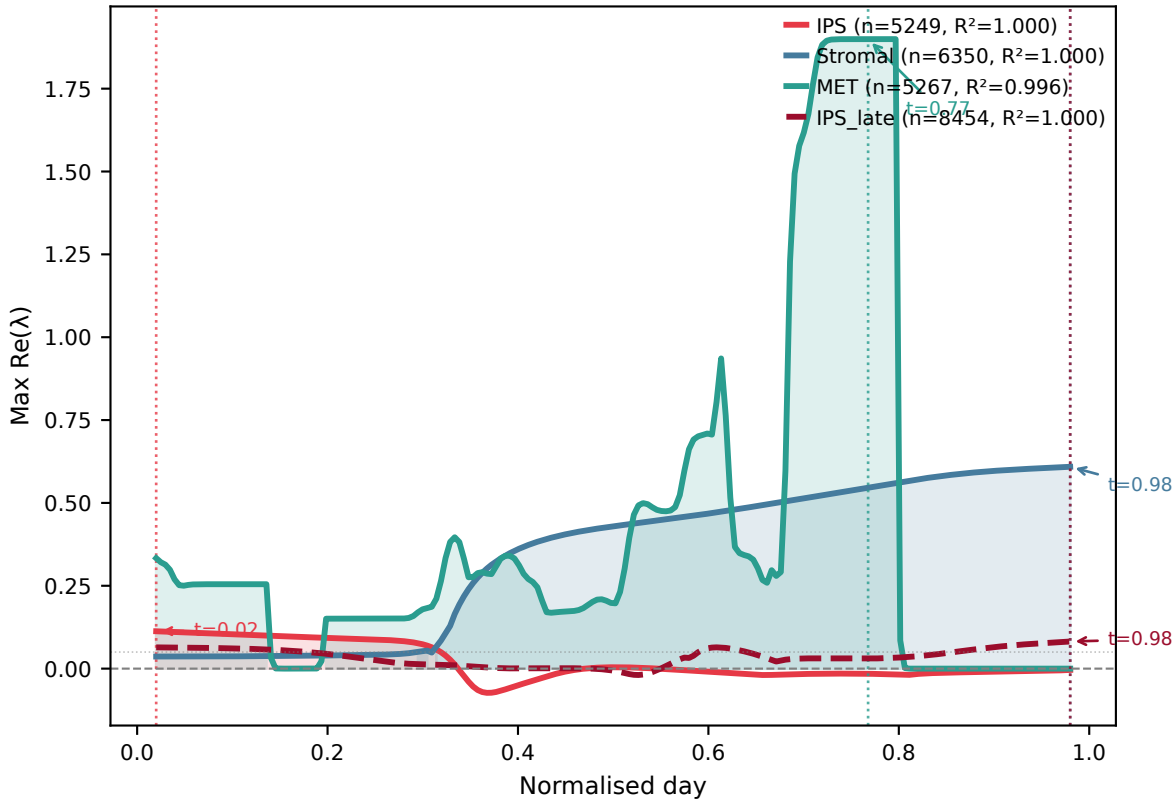
Figure 4 | MEF → iPSC Reprogramming (SCP295) — Branch-Separated scJDO

FactorAnalysis (30-D) + FLE + experimental day | panel c = IPS (full) vs IPS_late (day≥4) | IPS $R^2=1.000$ Stromal $R^2=1.000$ MET $R^2=0.996$ IPS_late $R^2=1.000$

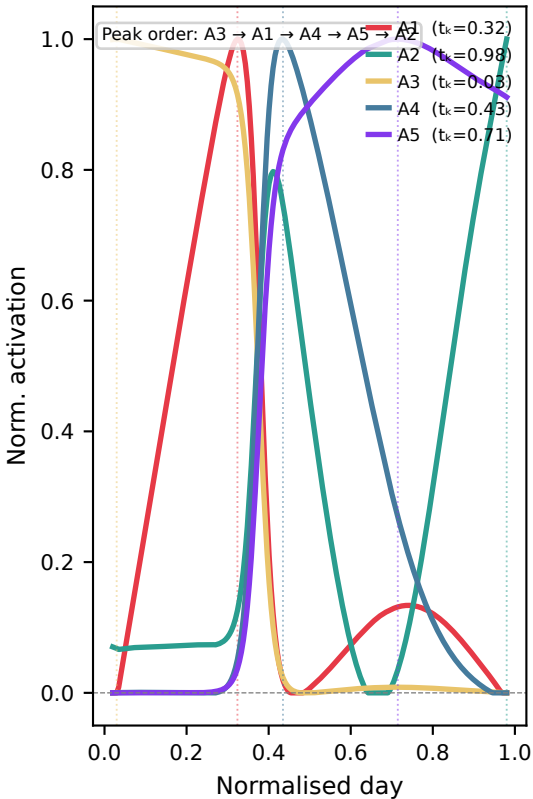
a | Reprogramming trajectory — SCP295 (24,999 cells, days 0-18)



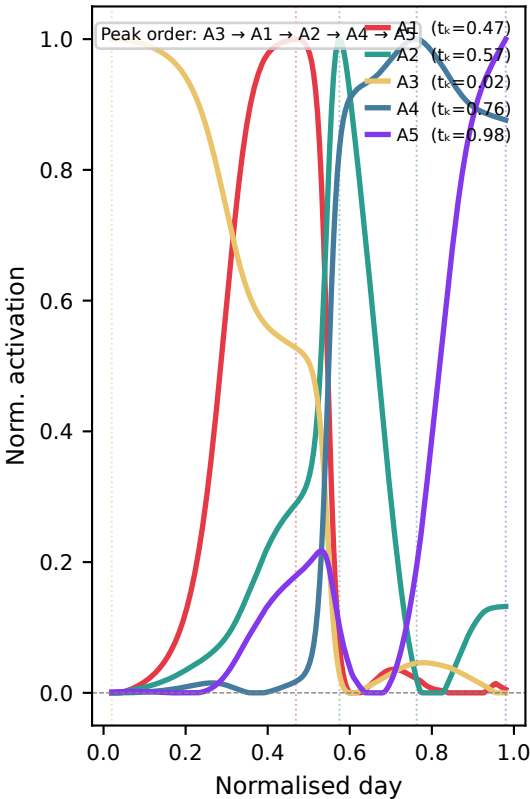
b | Local sensitivity — IPS / Stromal / MET + IPS_late



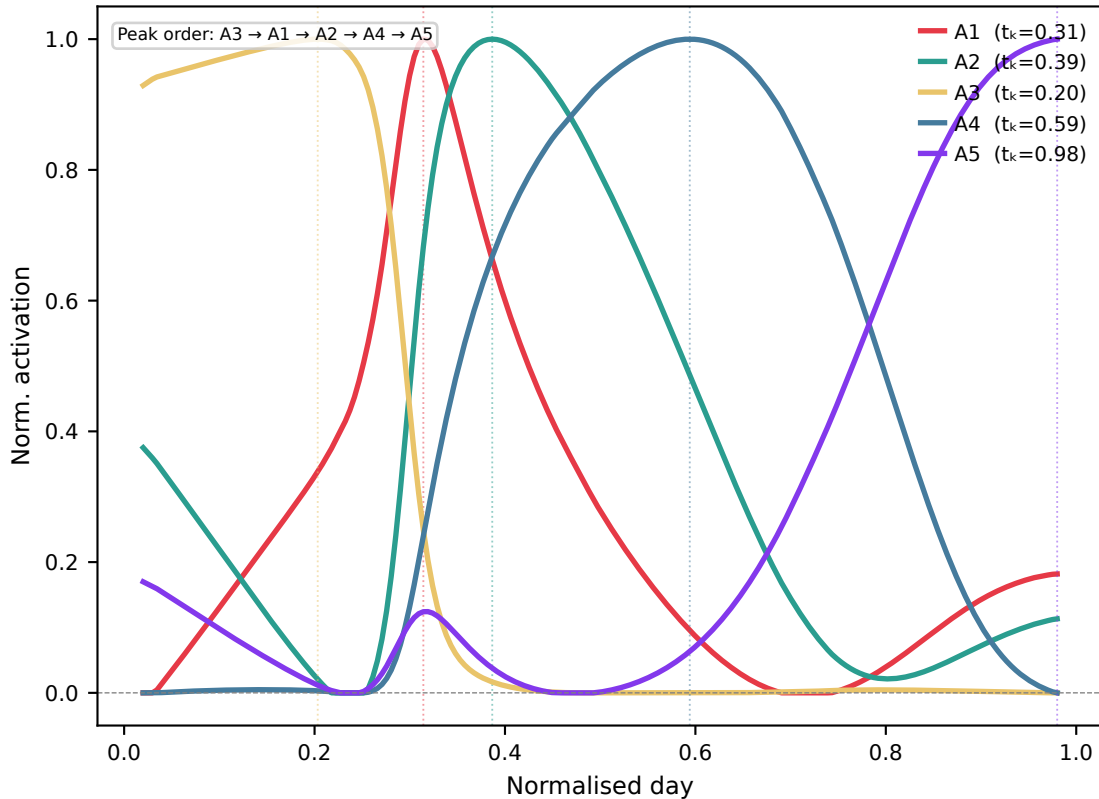
c | IPS (K=5, $R^2=1.000$, n=5249)



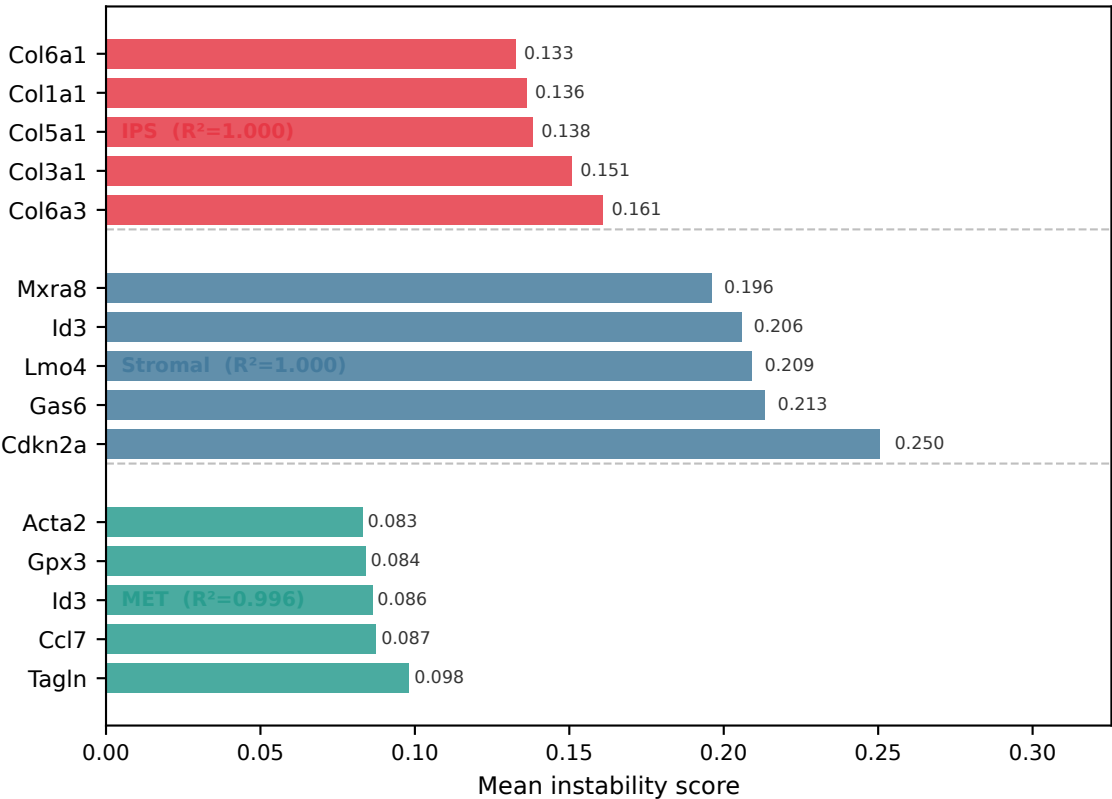
c' | IPS_late (K=5, $R^2=1.000$, n=8454)



d | Stromal (K=5, $R^2=1.000$, n=6350)



e | Top instability genes — IPS / Stromal / MET



f | MET (K=5, $R^2=0.996$, n=5267)

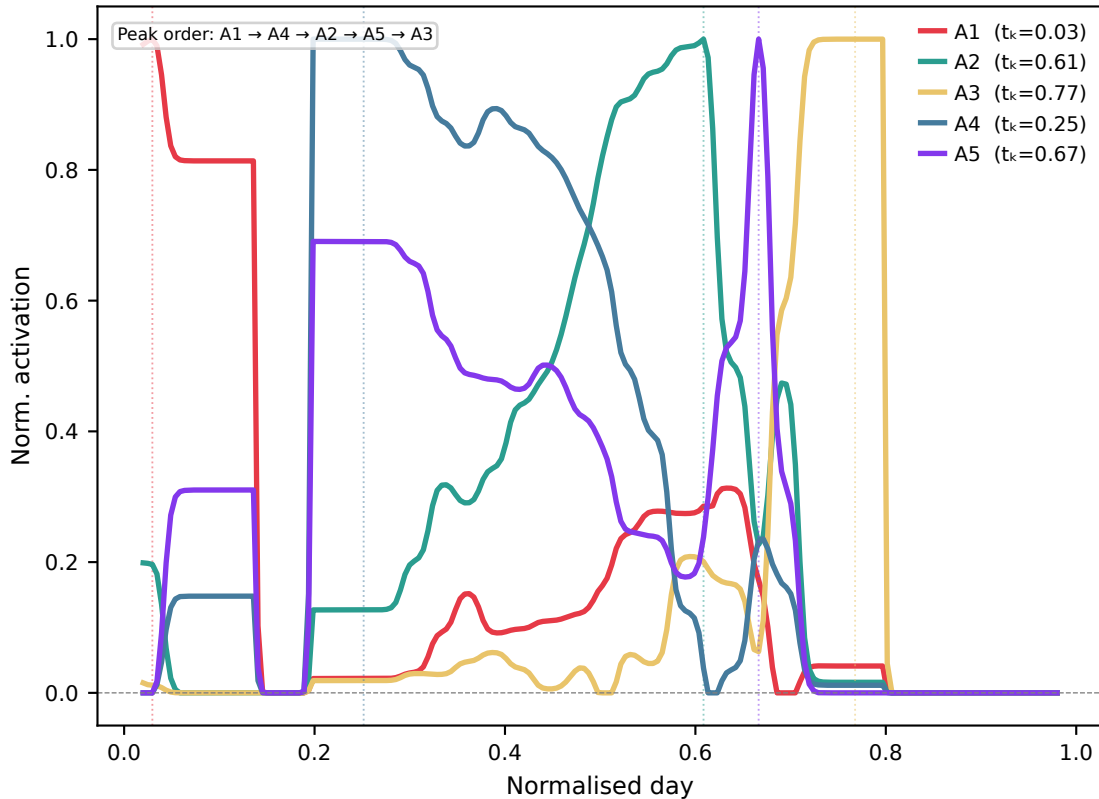
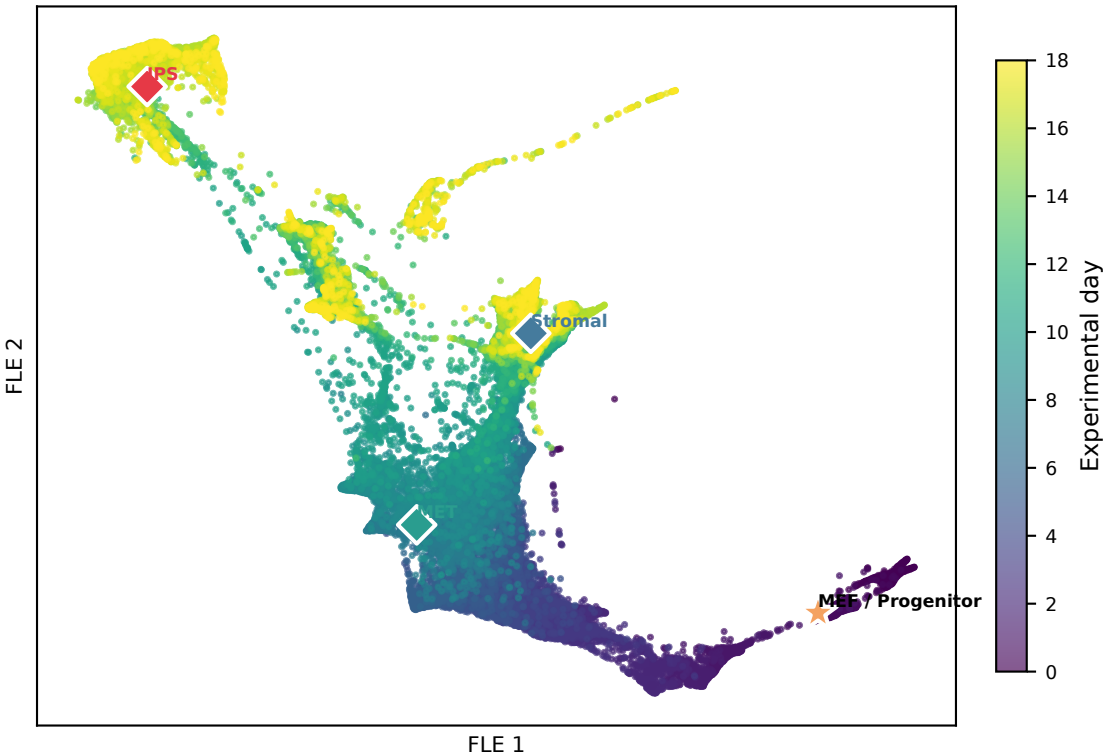


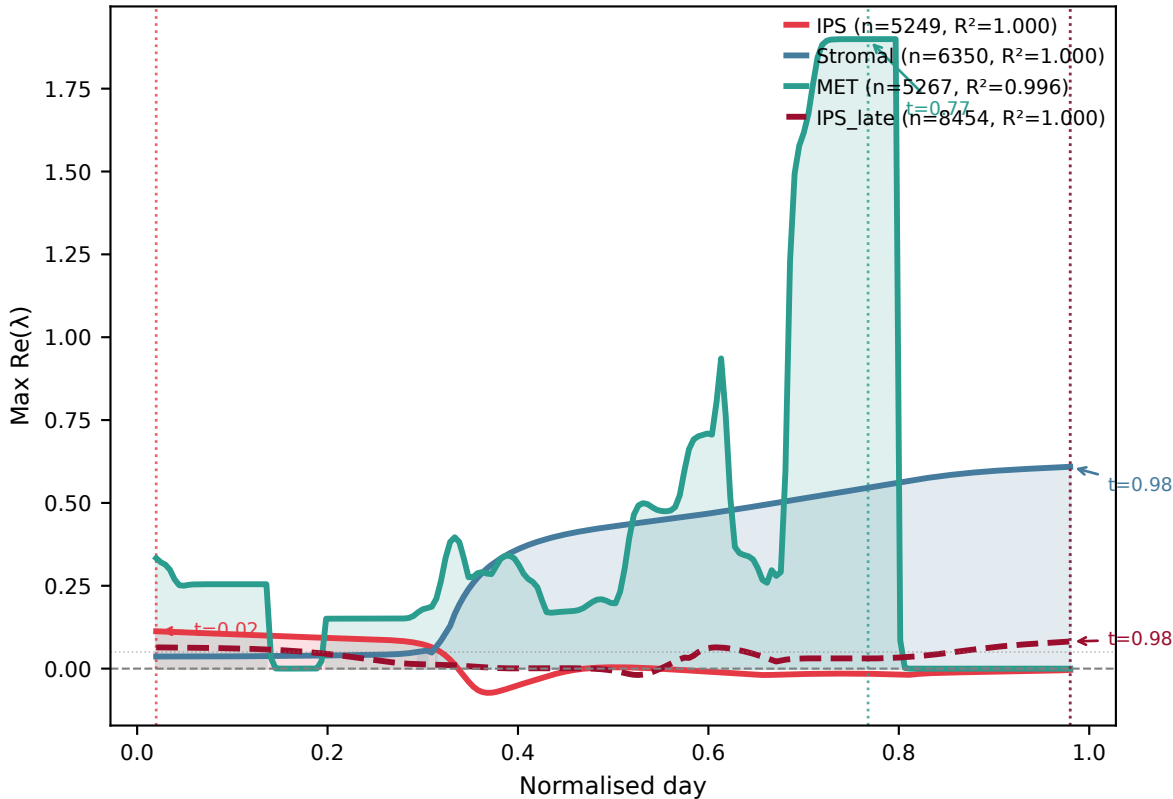
Figure 4 | MEF → iPSC Reprogramming (SCP295) — Branch-Separated scJDO

FactorAnalysis (30-D) + FLE + experimental day | panel c = IPS (full) vs IPS_late (day≥4) | IPS $R^2=1.000$ Stromal $R^2=1.000$ MET $R^2=0.996$ IPS_late $R^2=1.000$

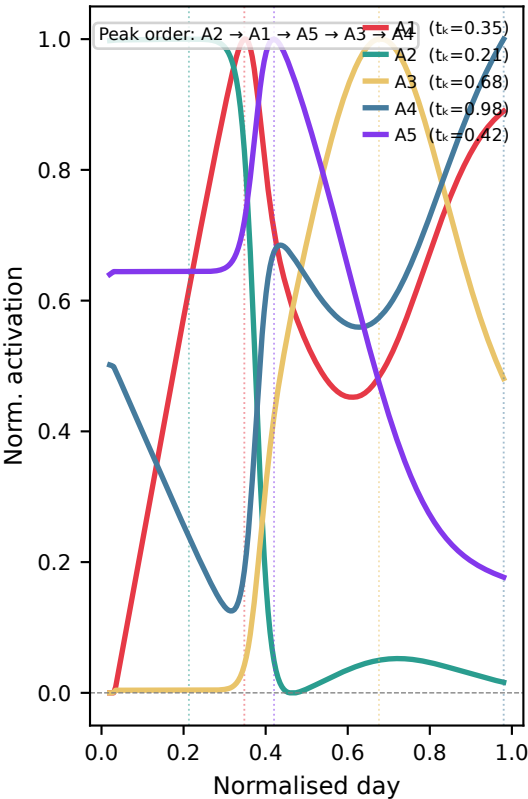
a | Reprogramming trajectory — SCP295 (24,999 cells, days 0-18)



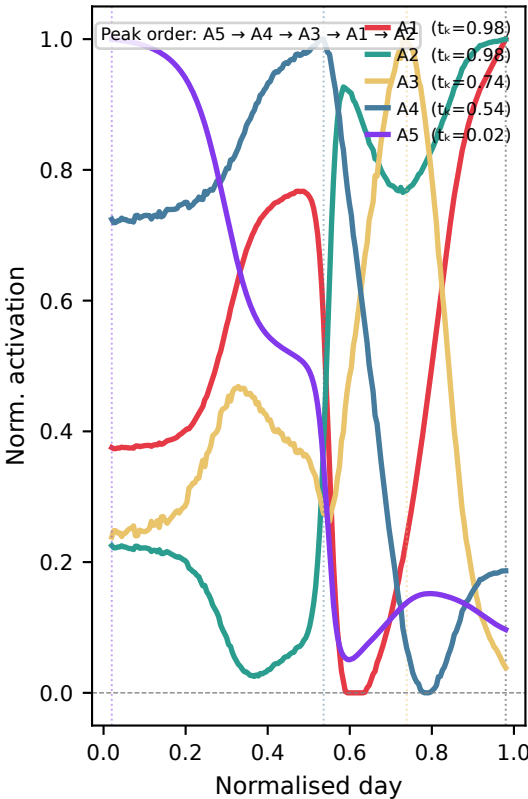
b | Local sensitivity — IPS / Stromal / MET + IPS_late



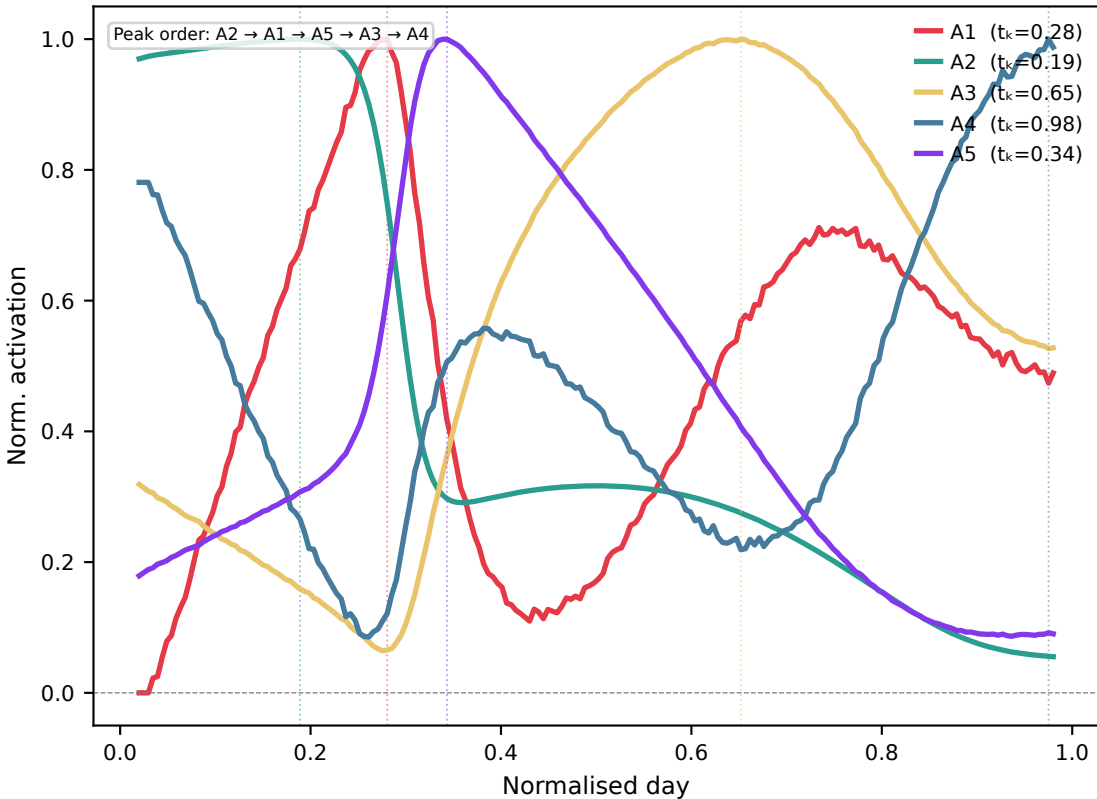
c | IPS (K=5, $R^2=1.000$, n=5249)



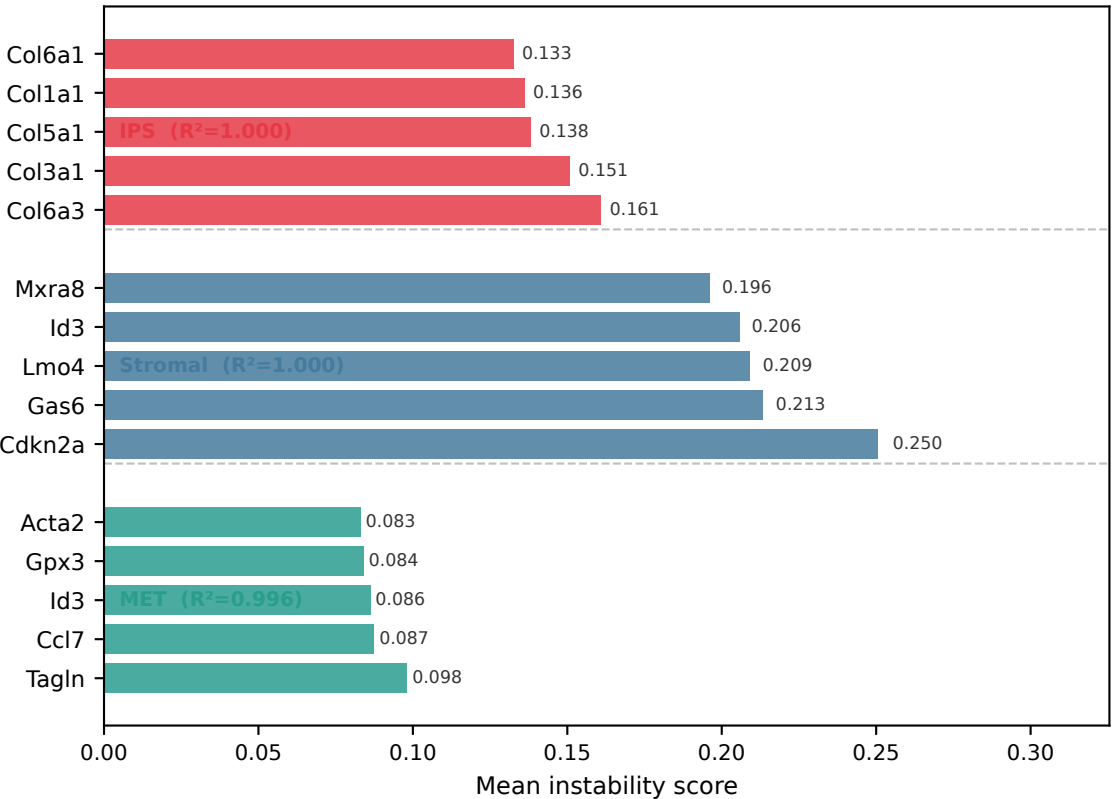
c' | IPS_late (K=5, $R^2=1.000$, n=8454)



d | Stromal (K=5, $R^2=1.000$, n=6350)



e | Top instability genes — IPS / Stromal / MET



f | MET (K=5, $R^2=0.996$, n=5267)

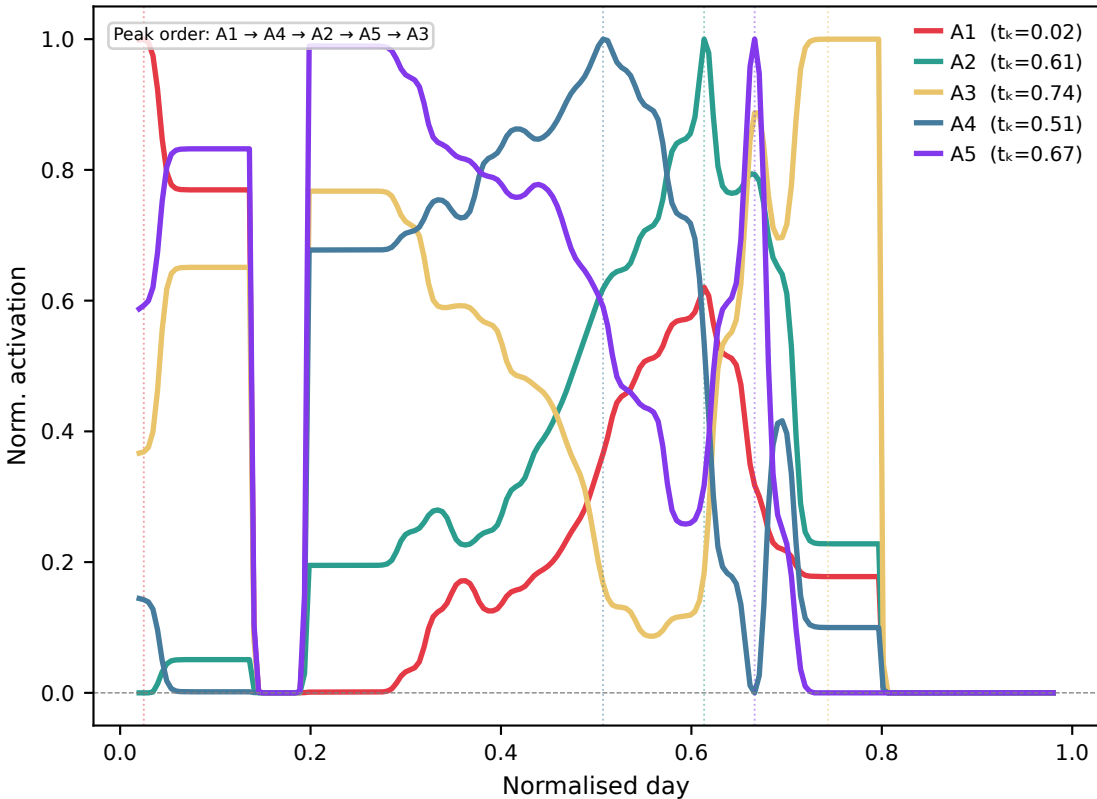
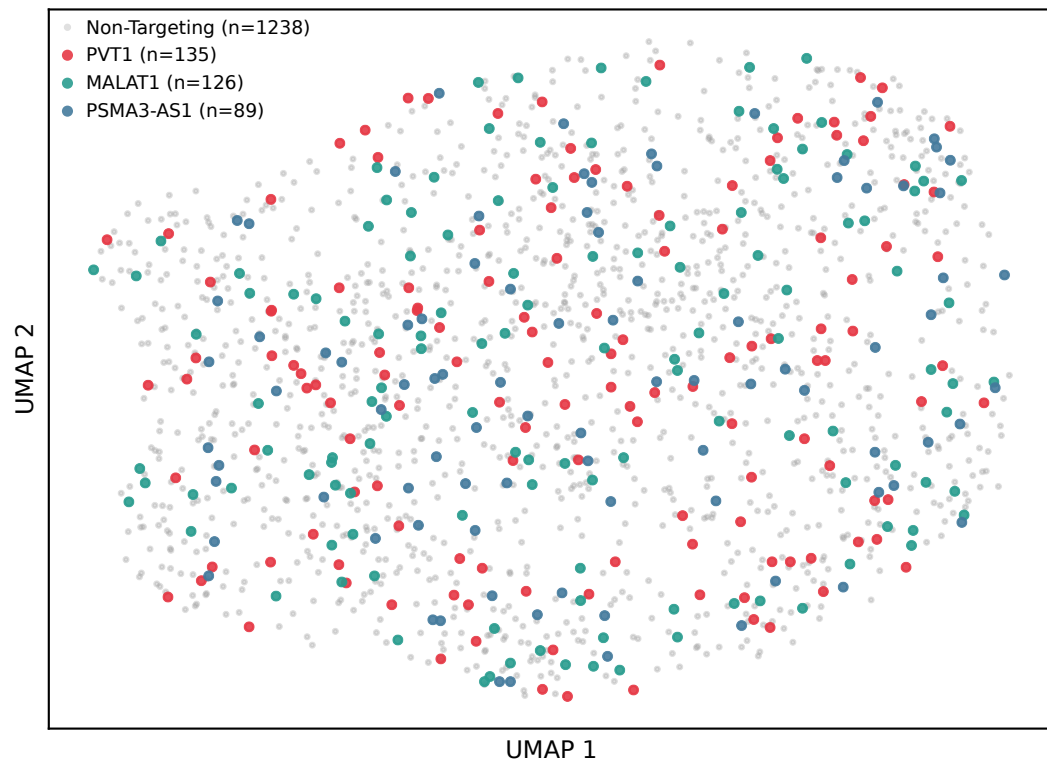
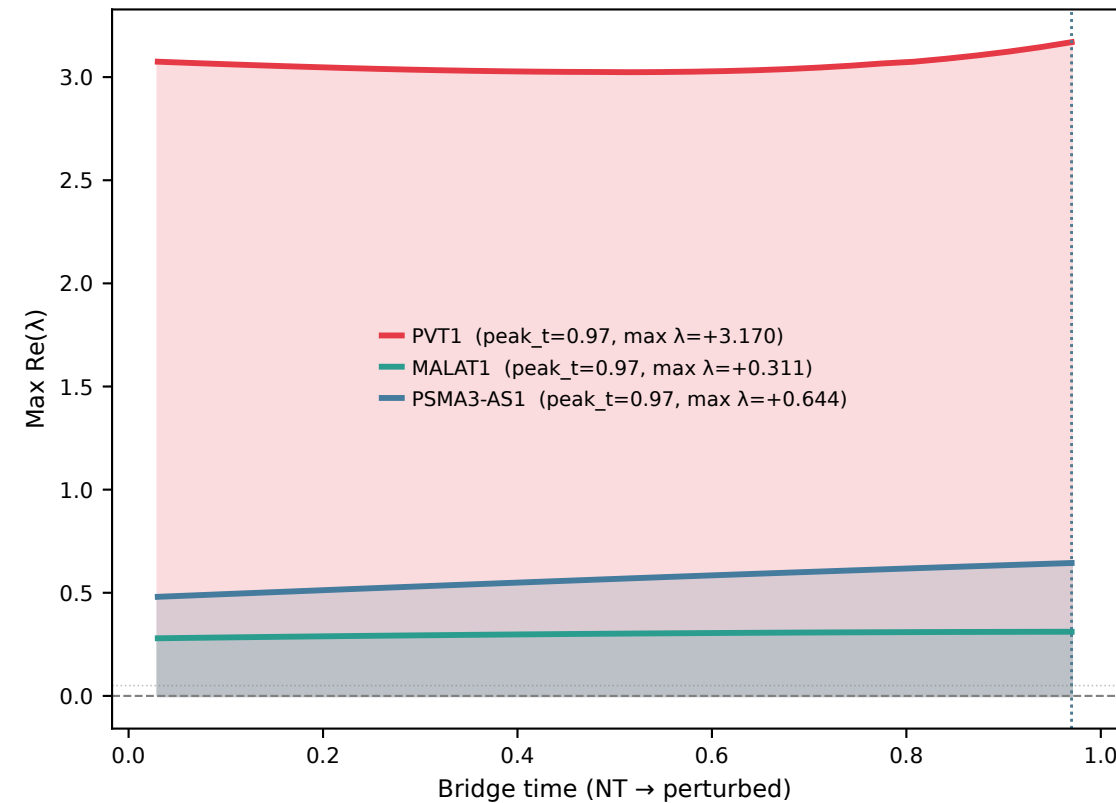


Figure 5 | K562 CRISPRi Perturb-seq — Schrödinger-bridge operator analysis
FactorAnalysis (30-D) + per-perturbation bridge (NT → target) | targets: PVT1, MALAT1, PSMA3-AS1

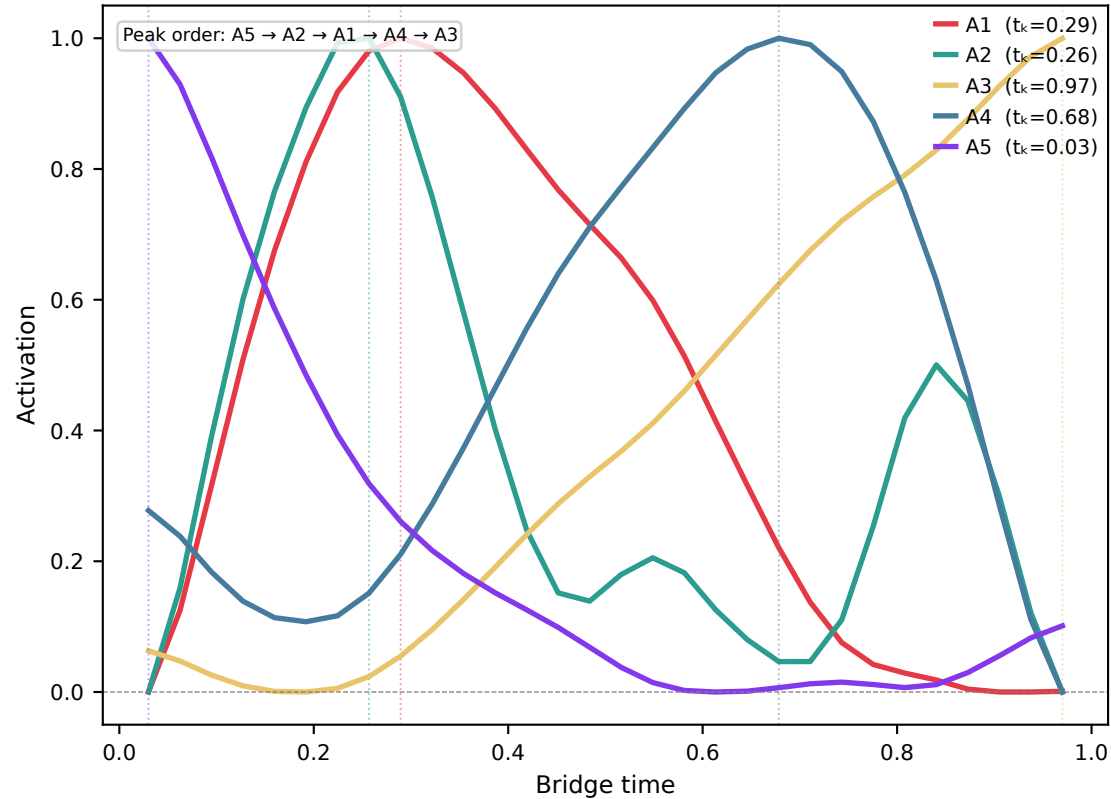
a | K562 Perturb-seq — NT + 3 lncRNA targets



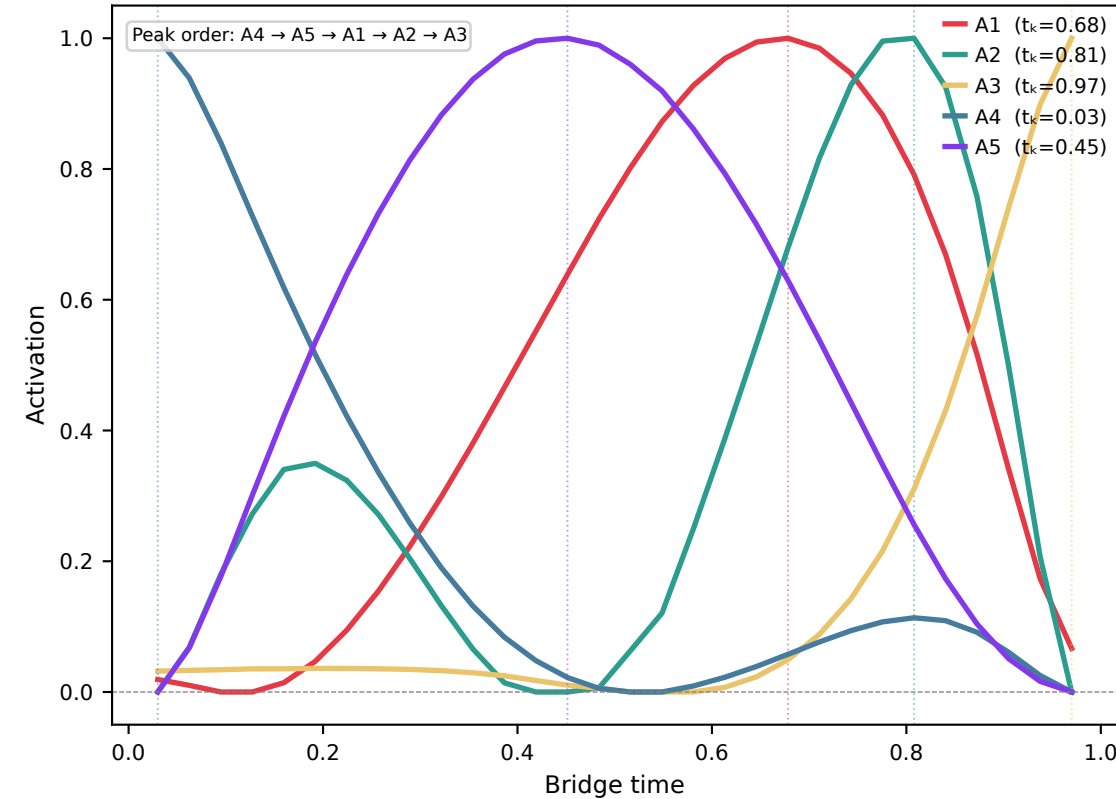
b | Forward bridge sensitivity — per perturbation



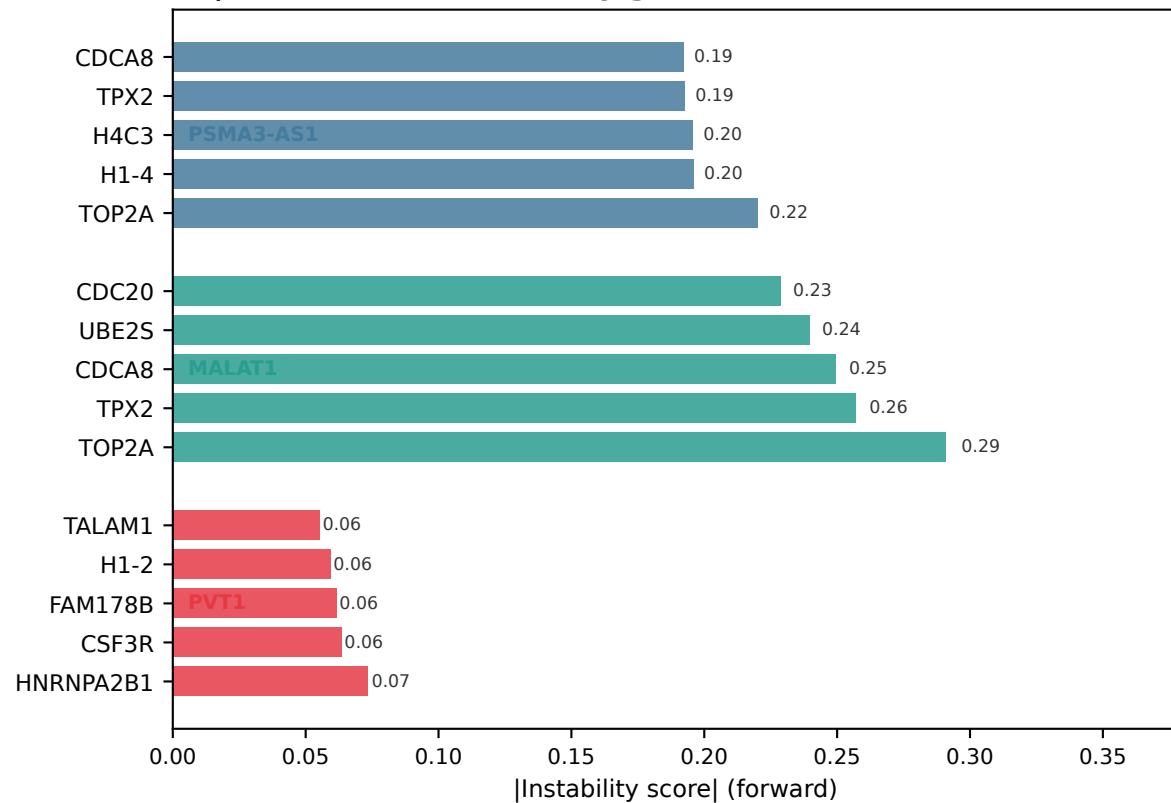
c | PVT1 archetypes (K=5)



d | MALAT1 archetypes (K=5)



e | Top forward instability genes per perturbation



f | PSMA3-AS1 archetypes (K=5)

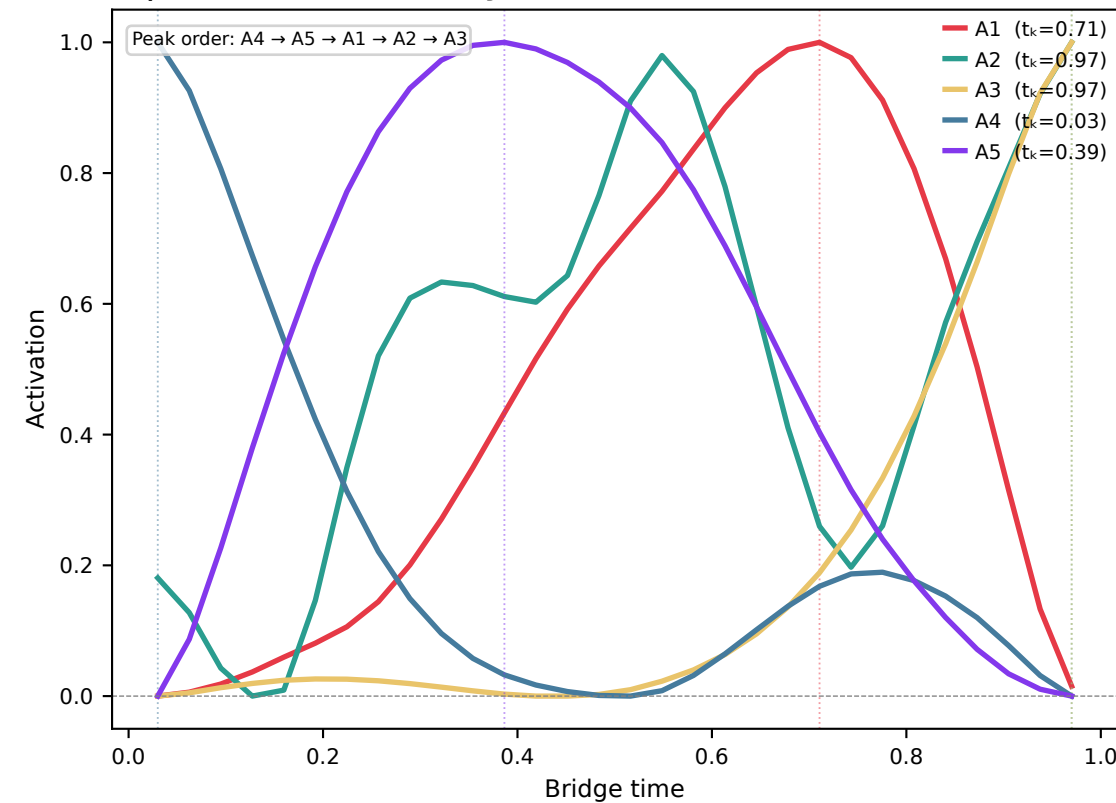
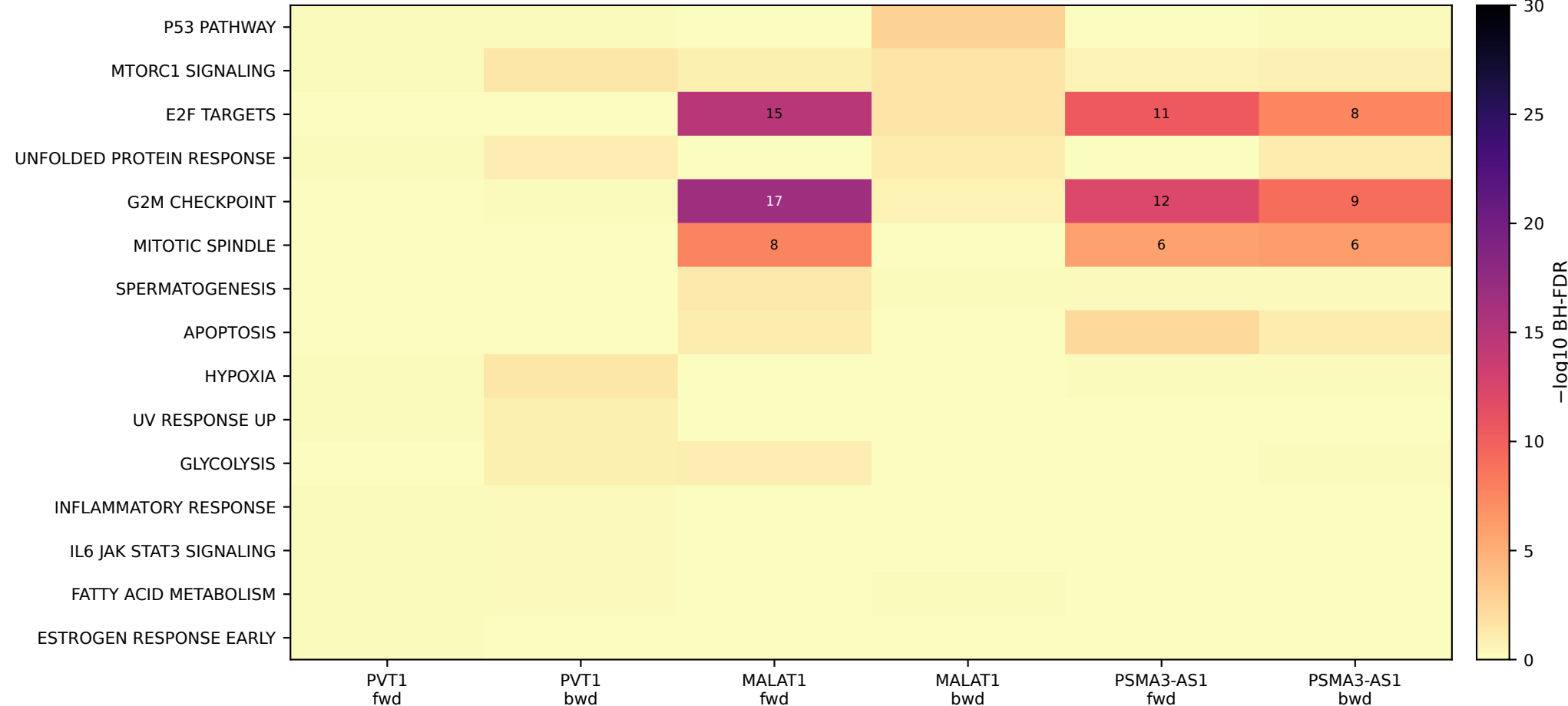


Figure 5 — Validation | Hallmark enrichment (gene programs) + permutation control (operator magnitude)

a | Hallmark over-representation in scJDO top-50 genes ($-\log_{10}$ FDR)



b | Permutation control (PVT1, n=20) empirical p = 0.143

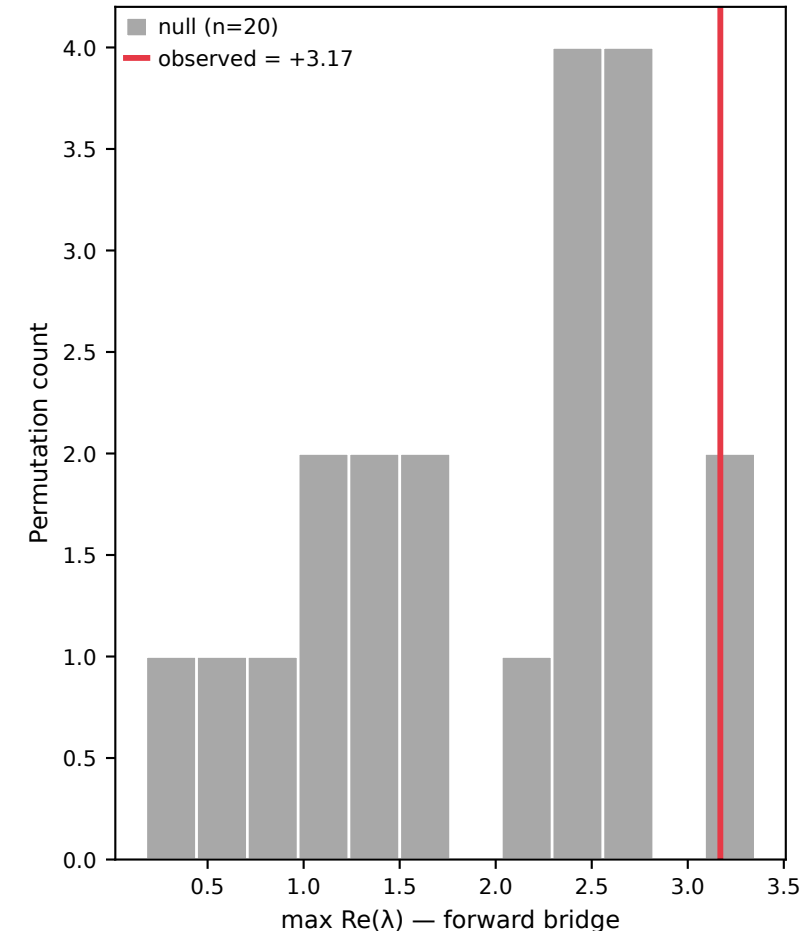
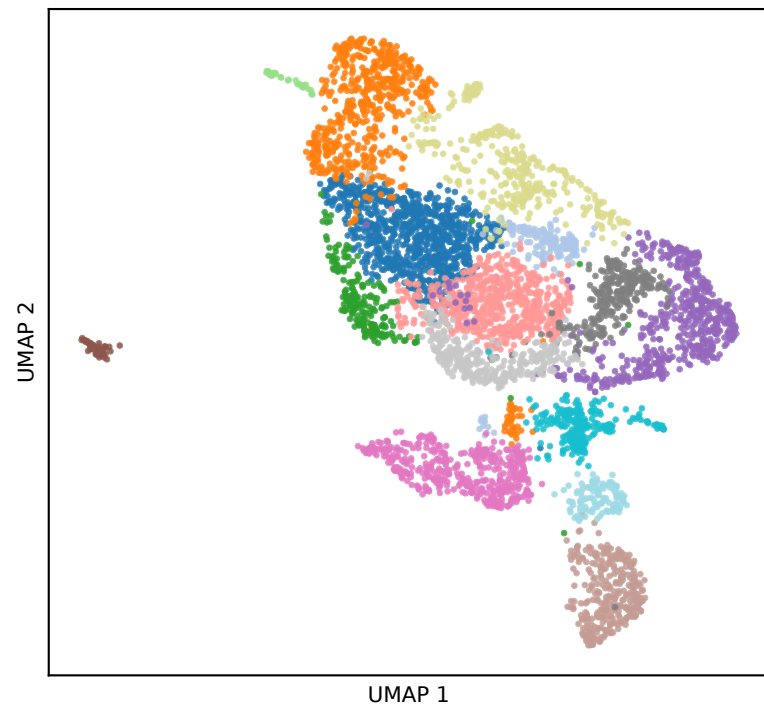
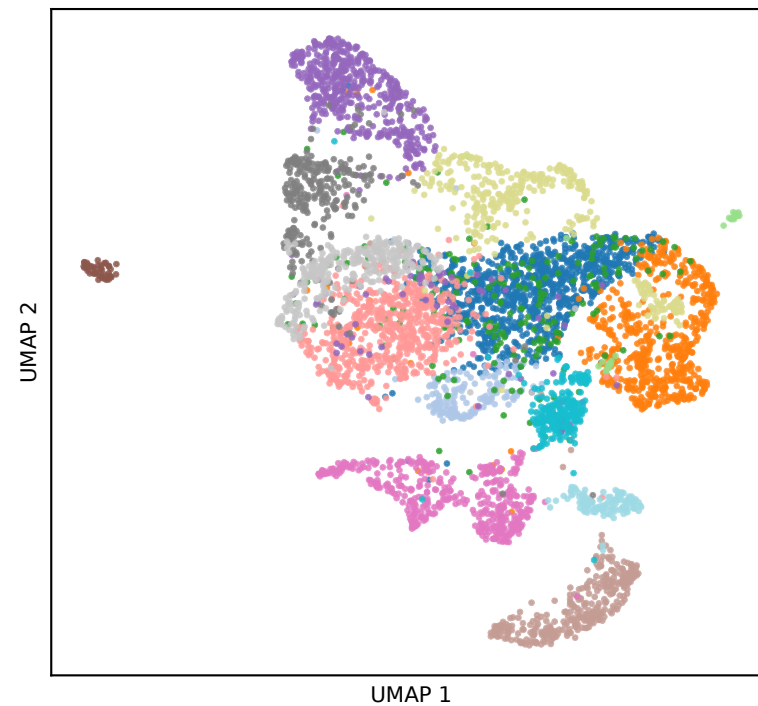


Figure 6 | Joint scRNA-seq + scATAC-seq Integration via FactorAnalysis
10x Multiome E18 mouse brain (4878 cells) · 2000 HVGs + 10000 variable peaks → FA(30)

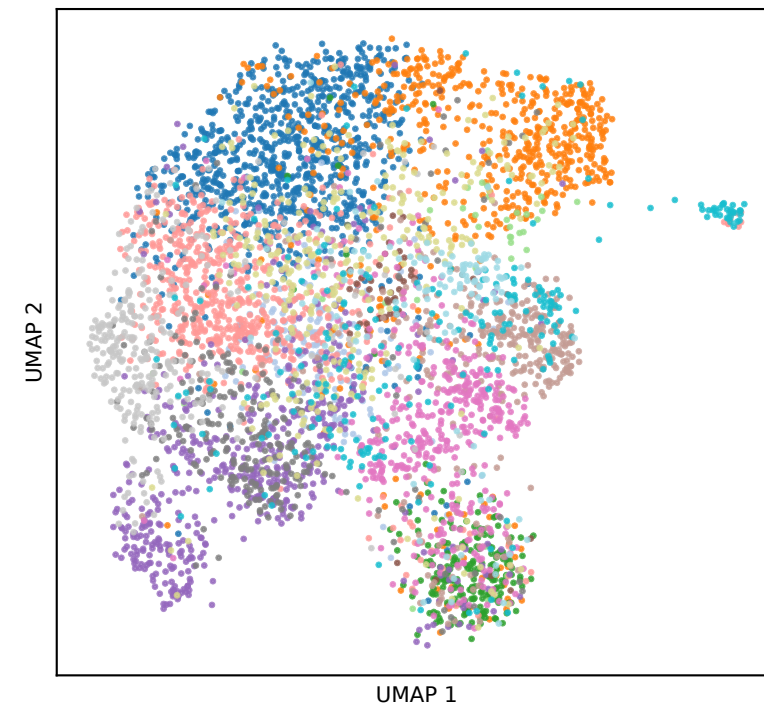
a | Joint FA UMAP (k=15)



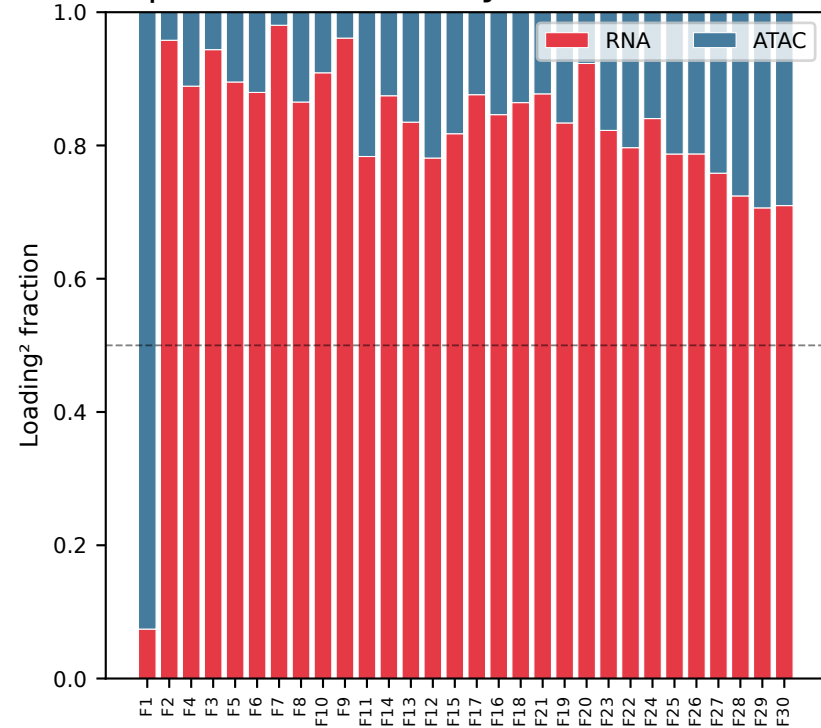
b | RNA-only FA UMAP (joint clusters)



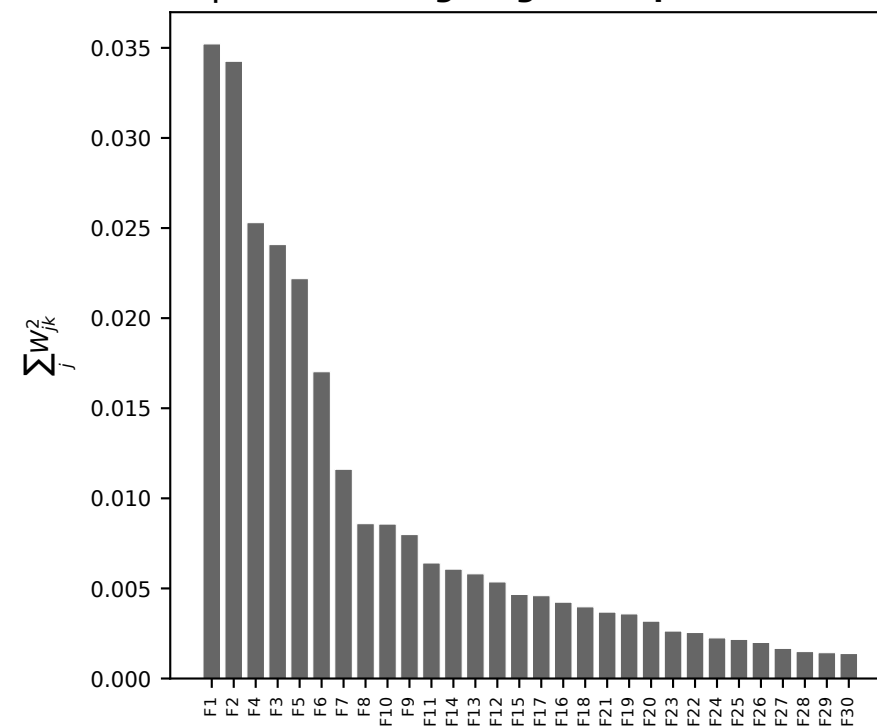
c | ATAC-only FA UMAP (joint clusters)



d | Per-factor modality contribution



e | Total loading magnitude per factor



f | Factor map: shared vs modality-specific

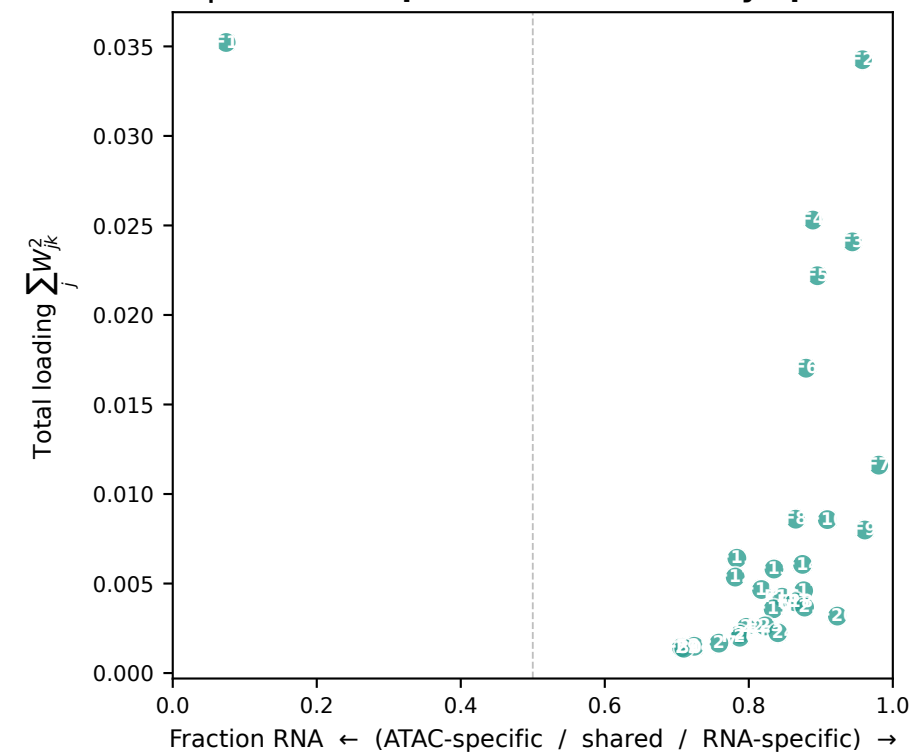
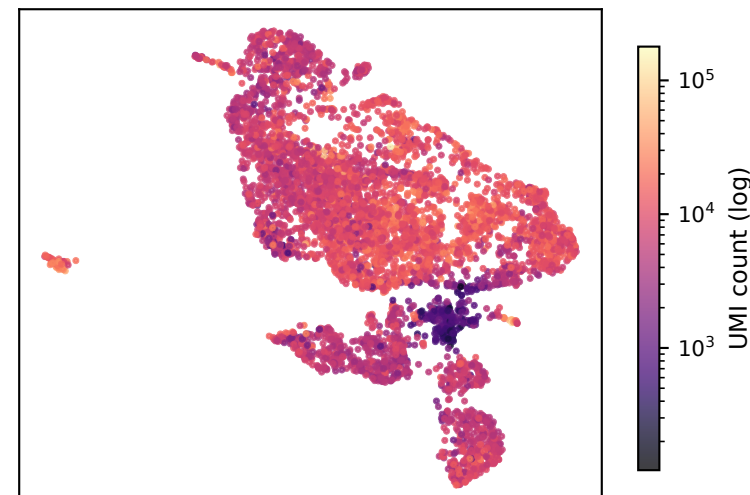


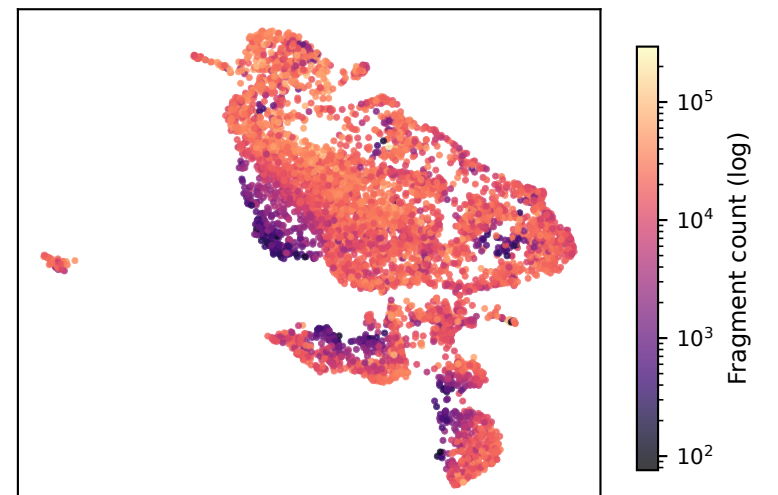
Figure 6 — Integration QC | joint scRNA + scATAC FA latent

10x Multiome E18 brain (4878 cells) · modality mixing · cluster preservation · kNN overlap · marker×promoter

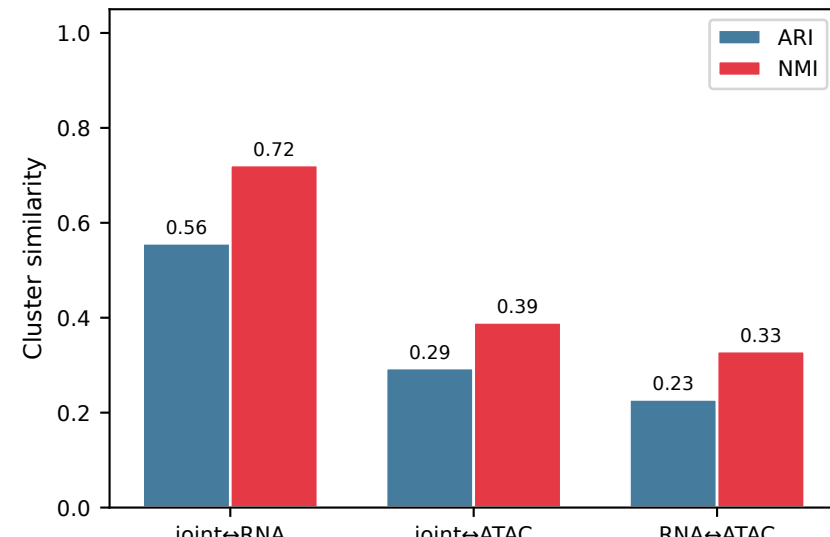
a | RNA UMIs (per cell)



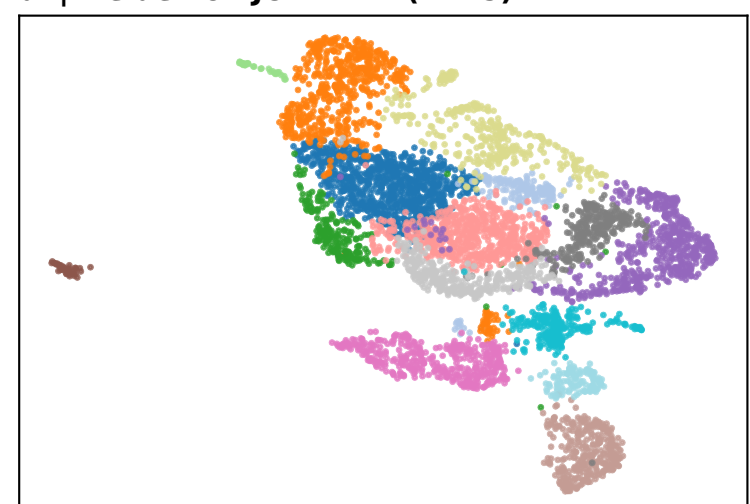
b | ATAC fragments (per cell)



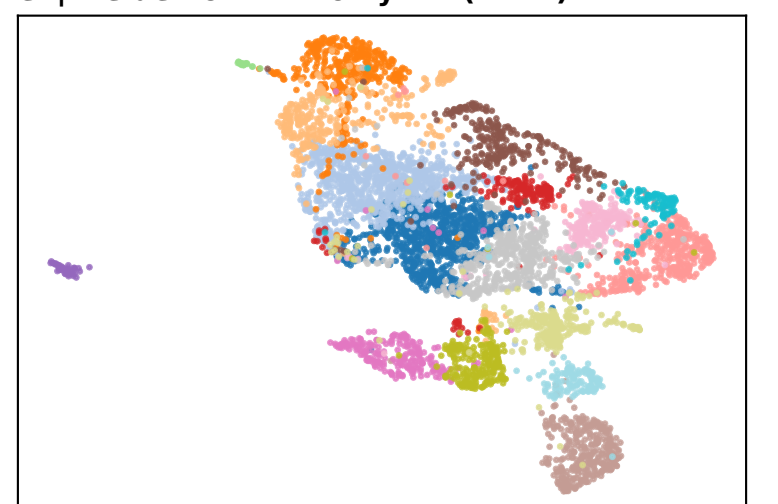
c | Cluster preservation (ARI/NMI)



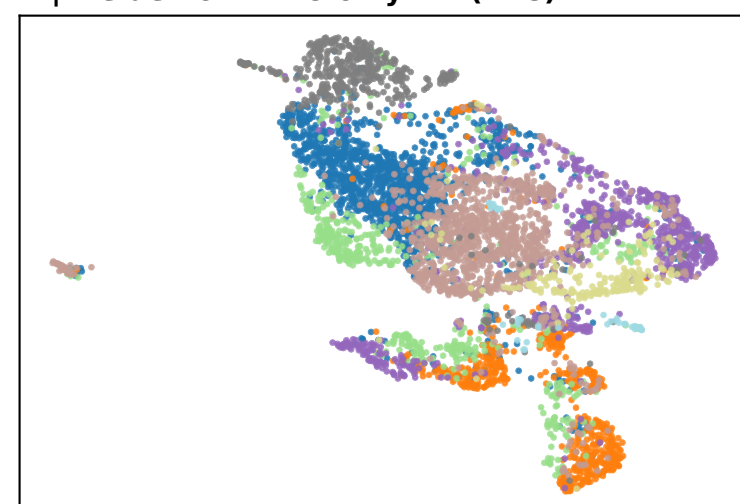
d | Leiden on JOINT FA (k=15)



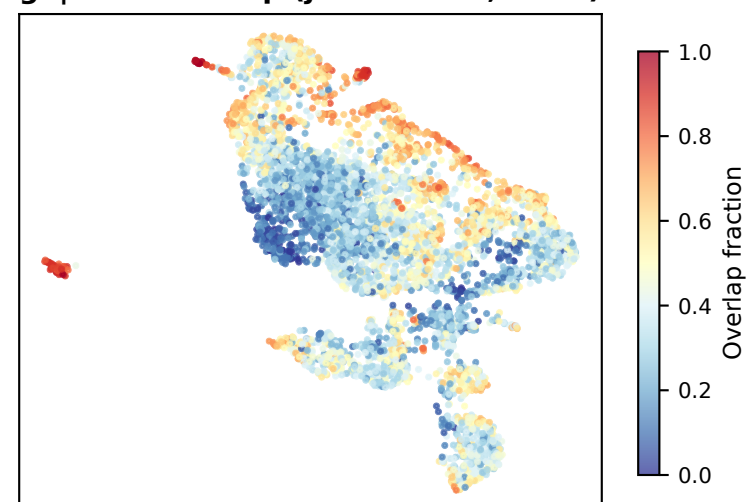
e | Leiden on RNA-only FA (k=17)



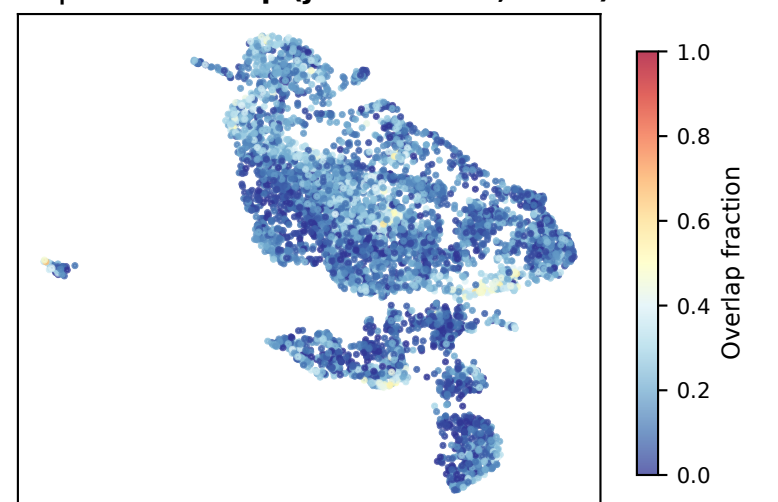
f | Leiden on ATAC-only FA (k=8)



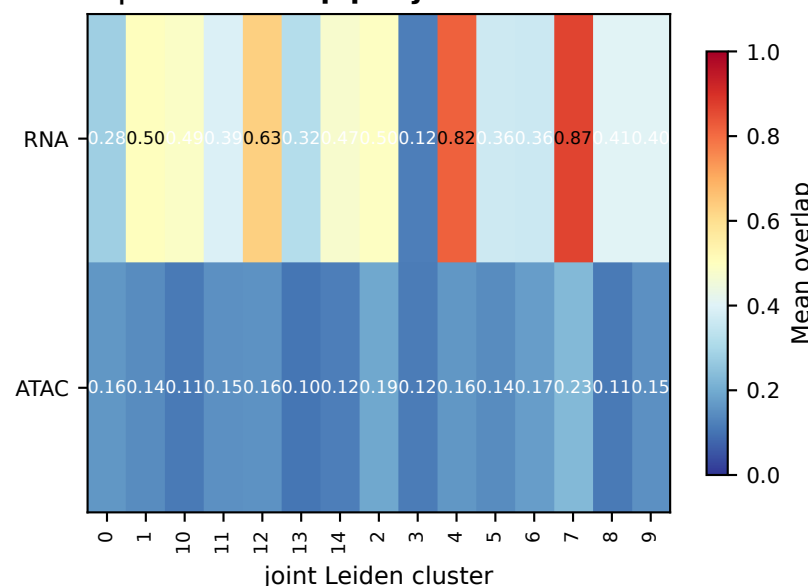
g | kNN overlap (joint ↔ RNA, k=30)



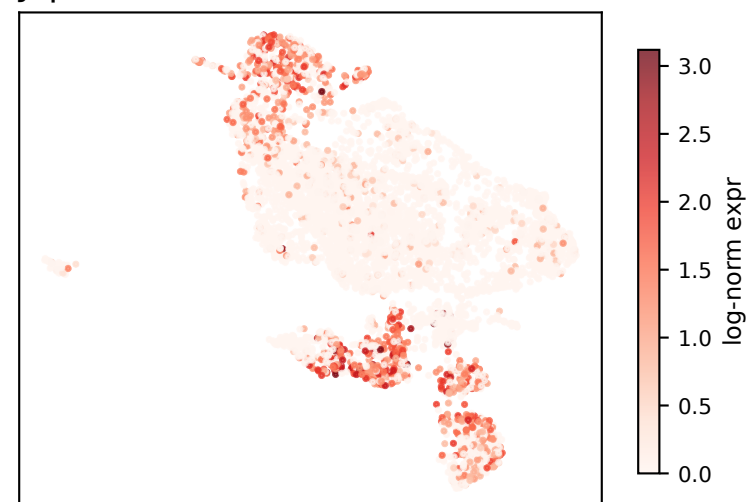
h | kNN overlap (joint ↔ ATAC, k=30)



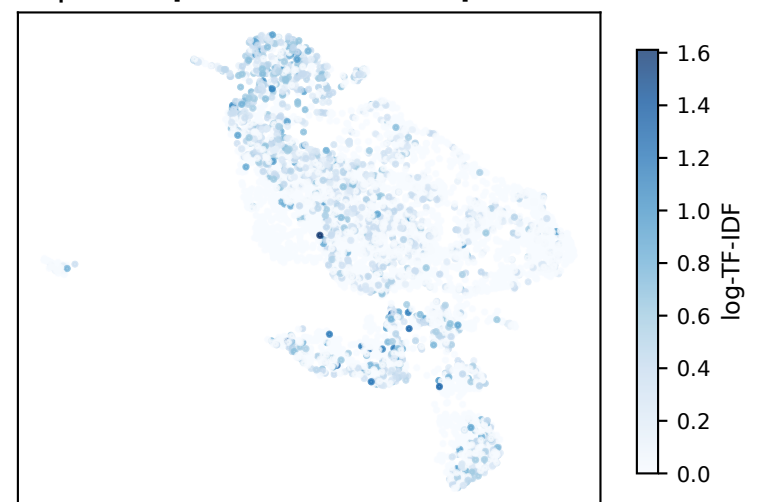
i | kNN overlap per joint cluster



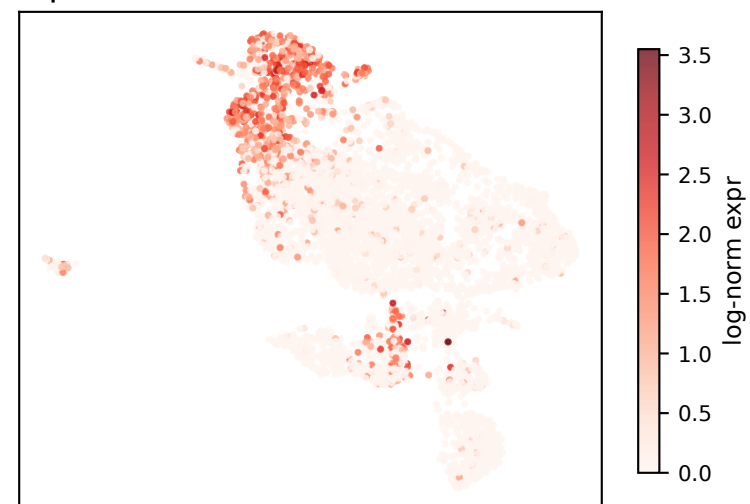
j | Sox2 RNA



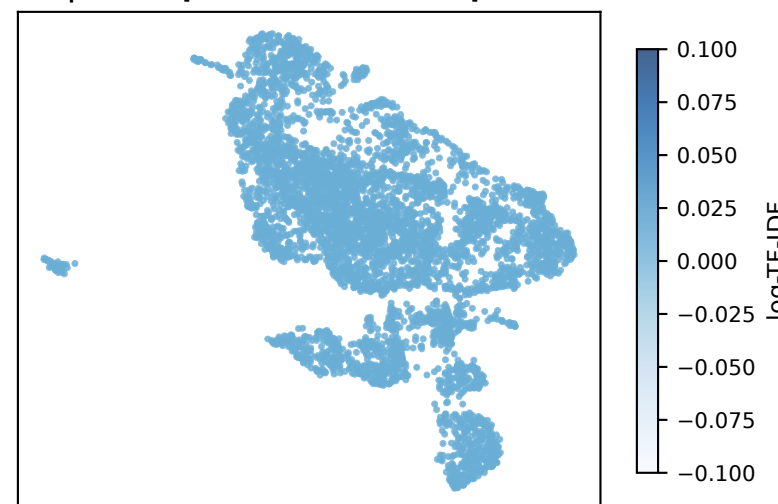
k | Sox2 promoter ATAC (3 pk)



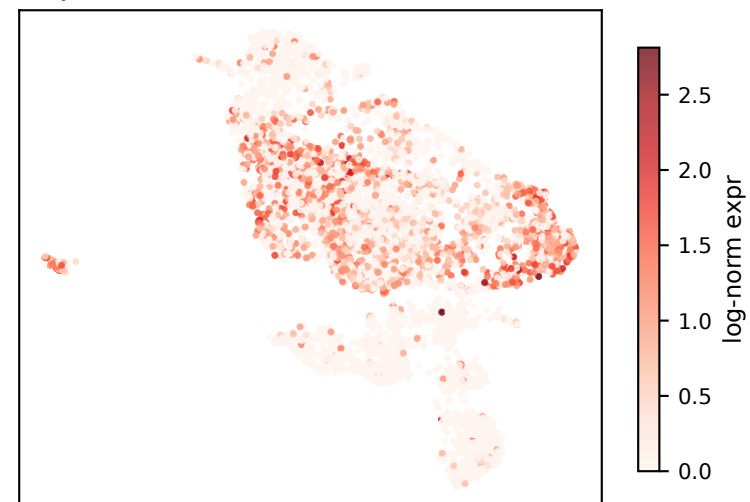
l | Pax6 RNA



m | Pax6 promoter ATAC (0 pk)



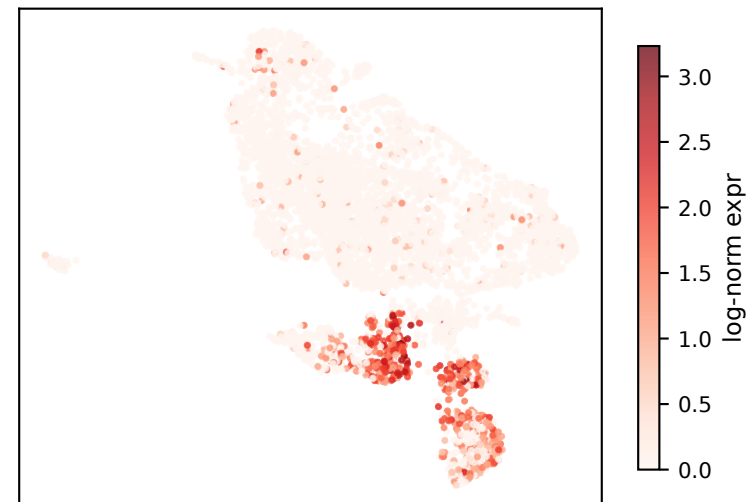
n | Tbr1 RNA



o | Tbr1 promoter ATAC (3 pk)



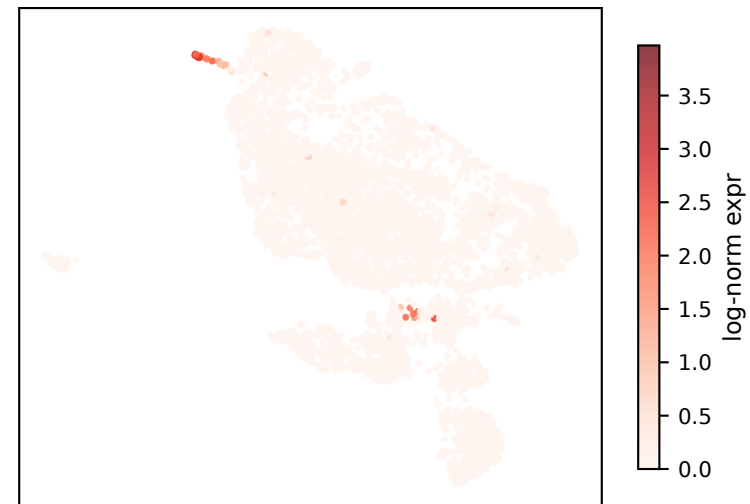
p | Dlx1 RNA



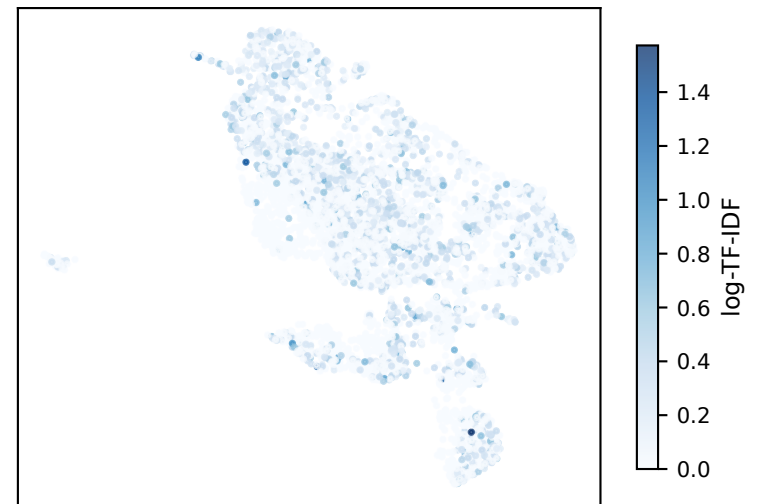
q | Dlx1 promoter ATAC (2 pk)



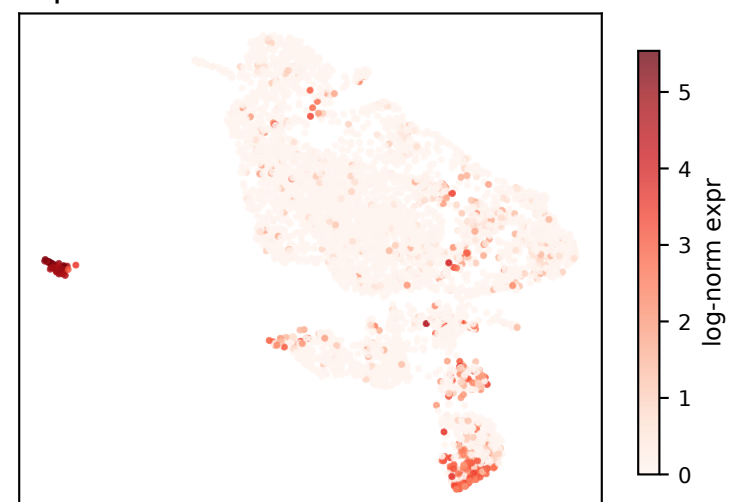
r | Pdgfra RNA



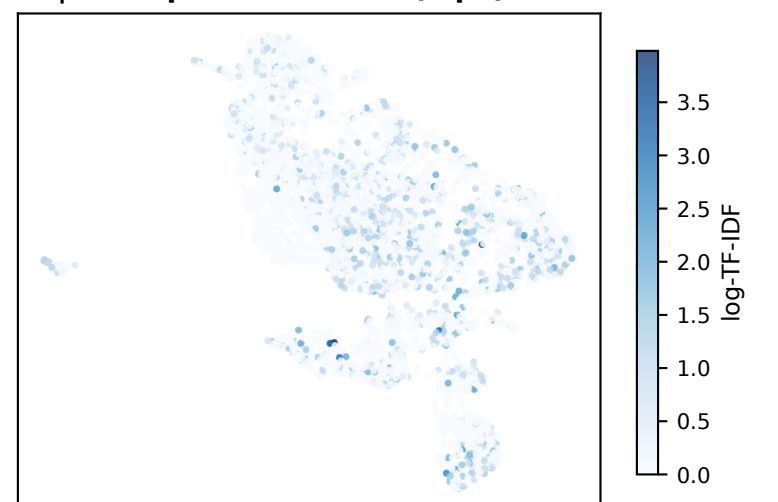
s | Pdgfra promoter ATAC (4 pk)



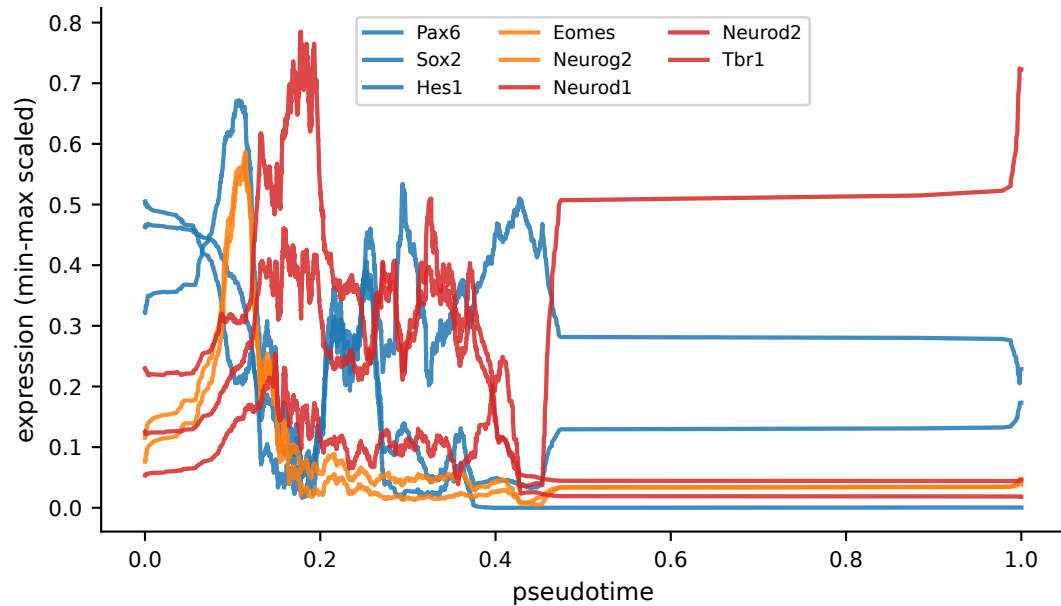
t | Reln RNA



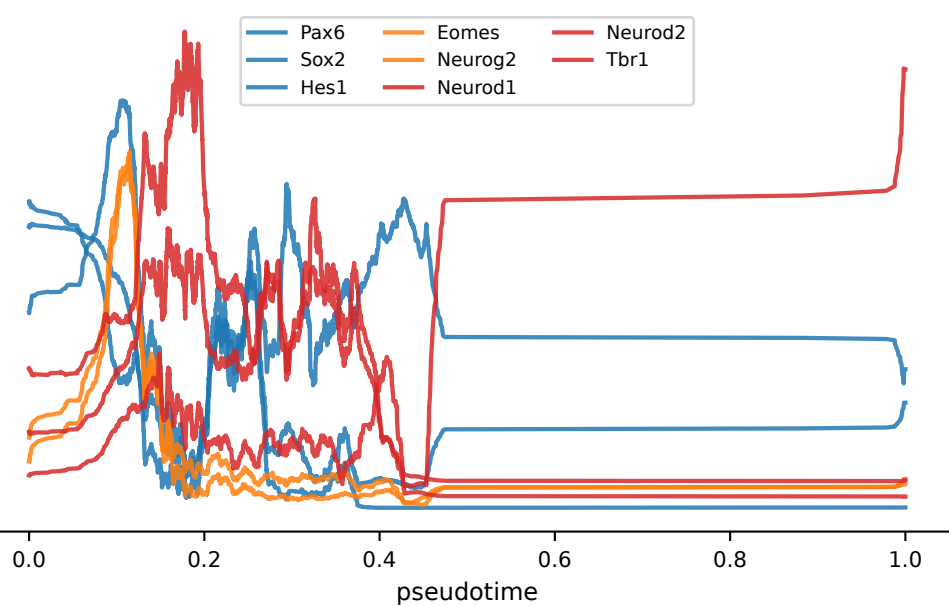
u | Reln promoter ATAC (1 pk)



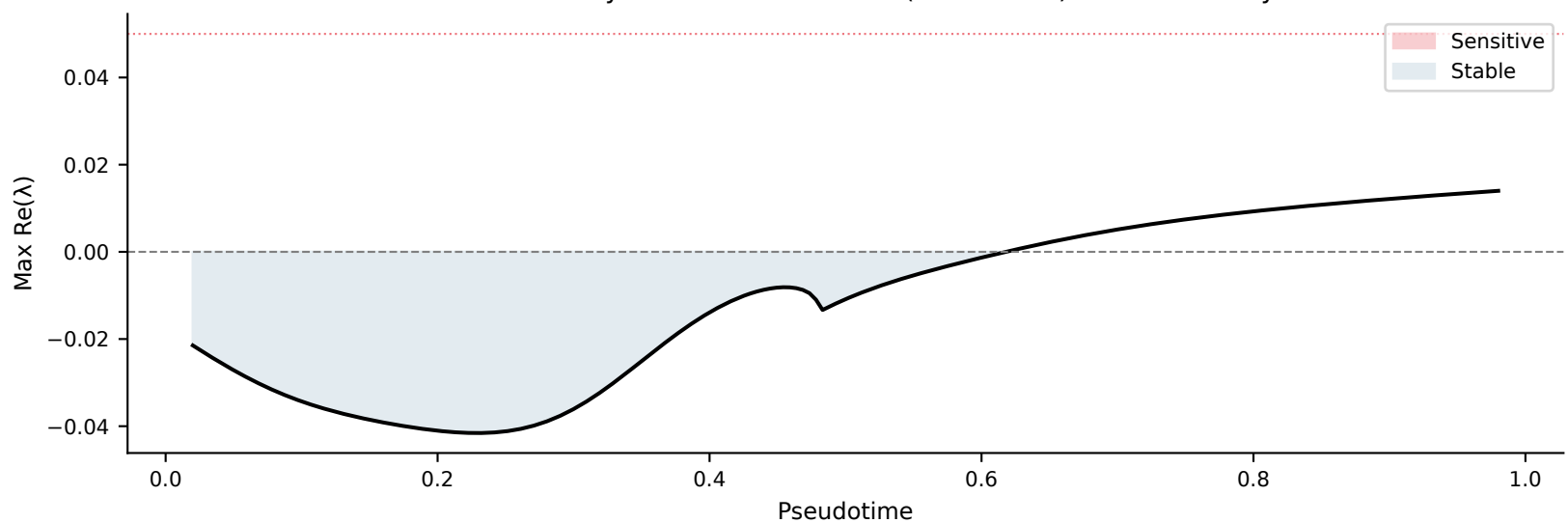
Uncorrected pseudotime



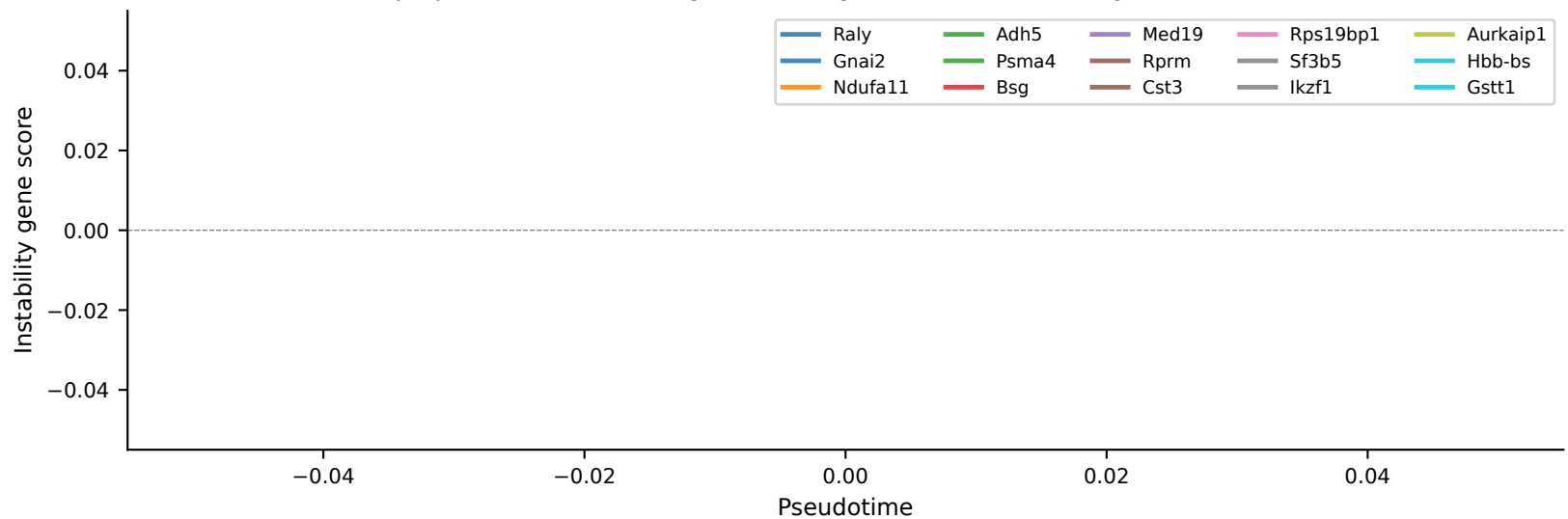
CC-regressed pseudotime



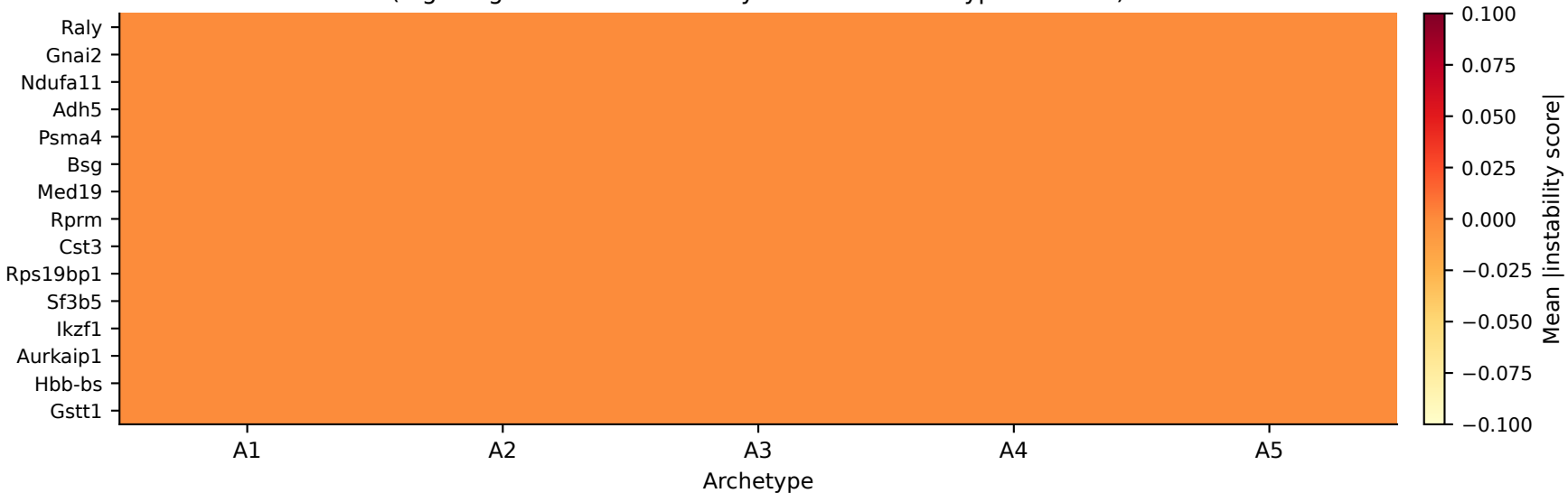
Local sensitivity — sensitive windows (shaded red) drive instability



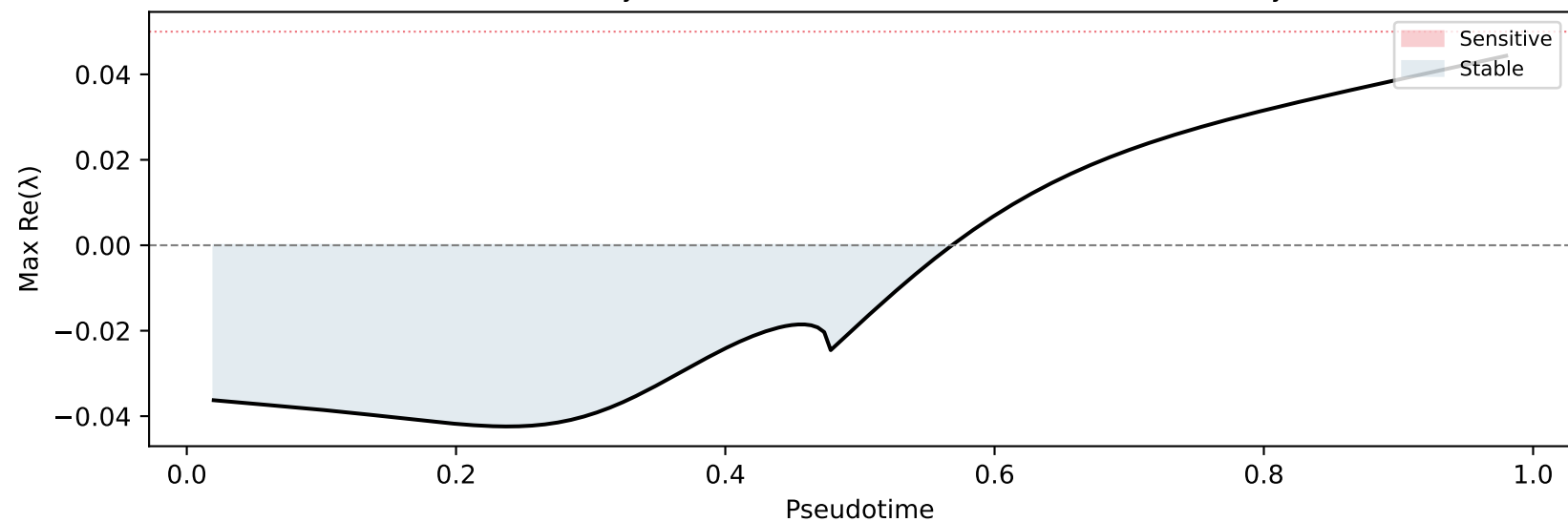
Top 15 instability-driving genes across pseudotime
(score = projection onto max-eigenvalue eigenvector; shown only in sensitive windows)



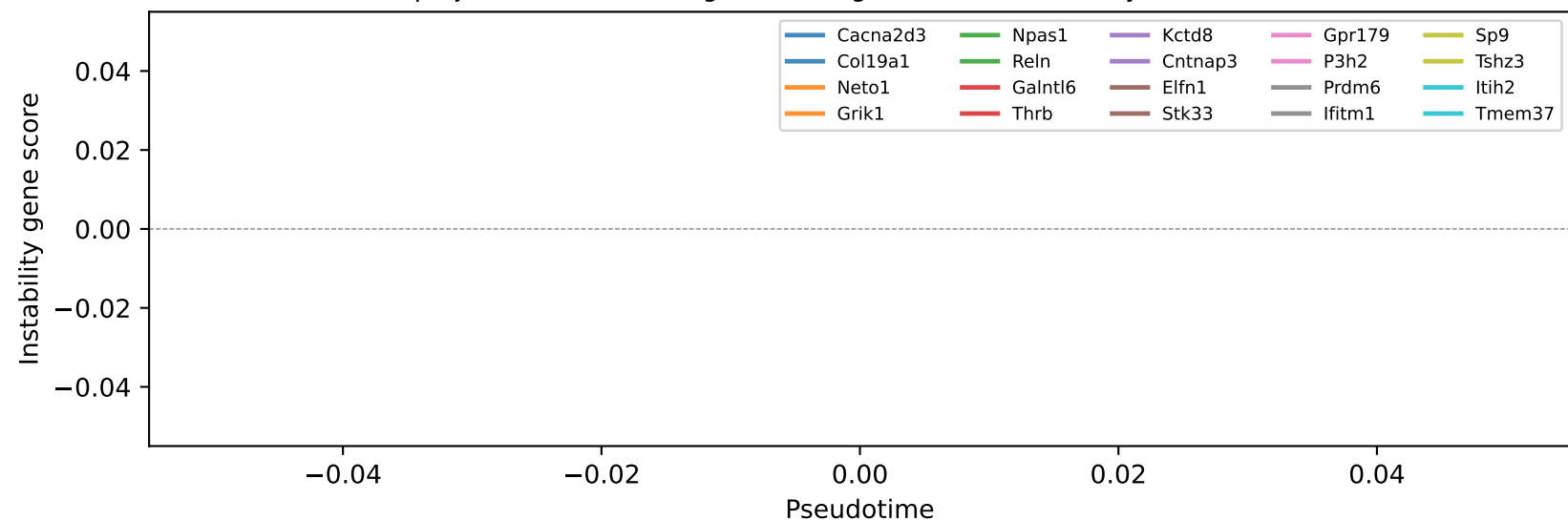
Mean instability score per gene × archetype
(high = gene drives instability when this archetype is active)



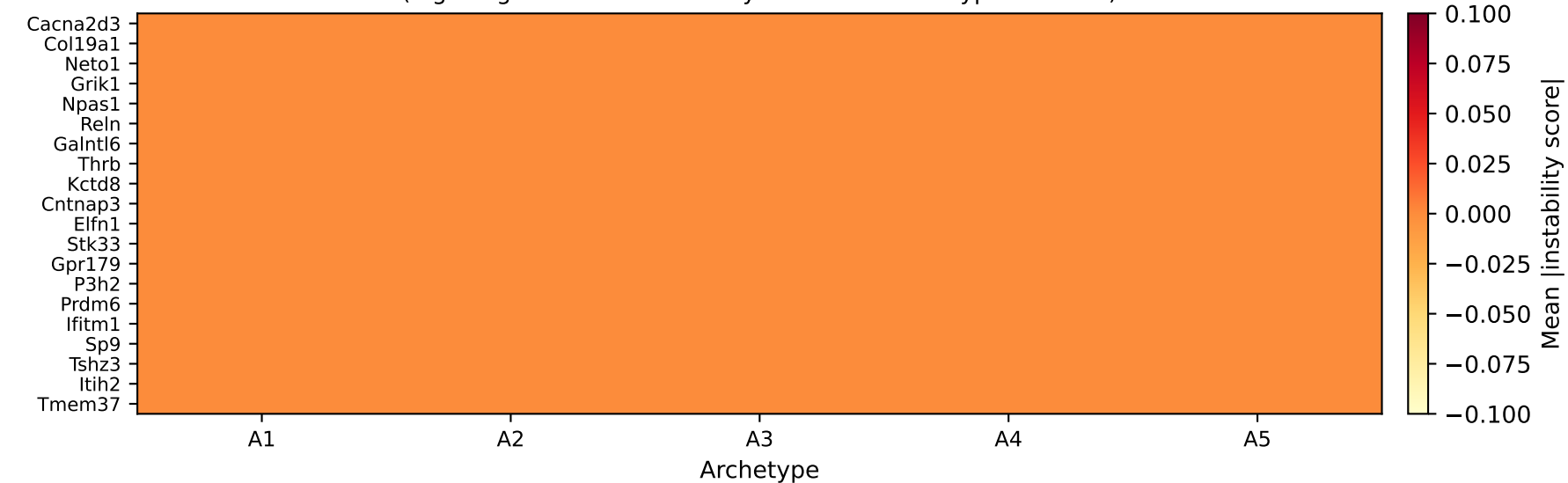
Local sensitivity — sensitive windows (shaded red) drive instability



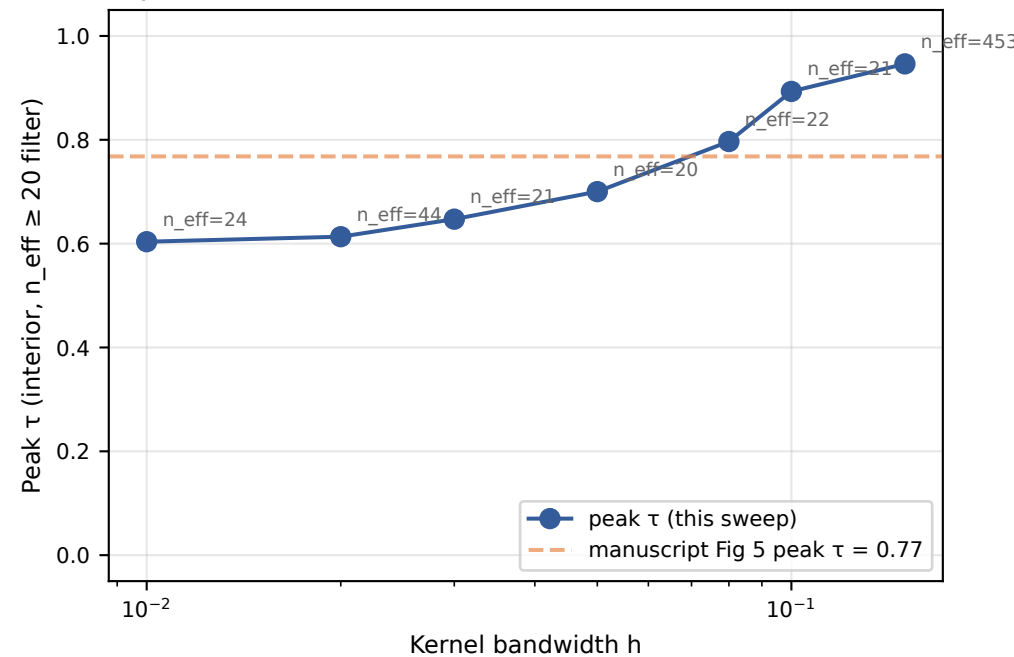
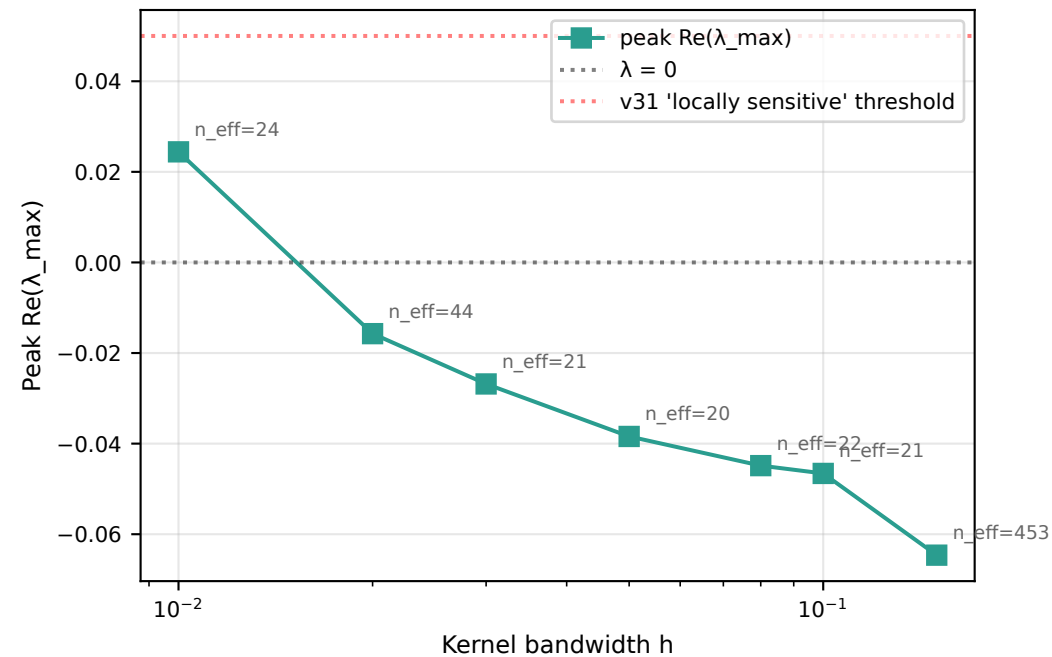
Top 20 instability-driving genes across pseudotime
(score = projection onto max-eigenvalue eigenvector; shown only in sensitive windows)



Mean instability score per gene × archetype
(high = gene drives instability when this archetype is active)

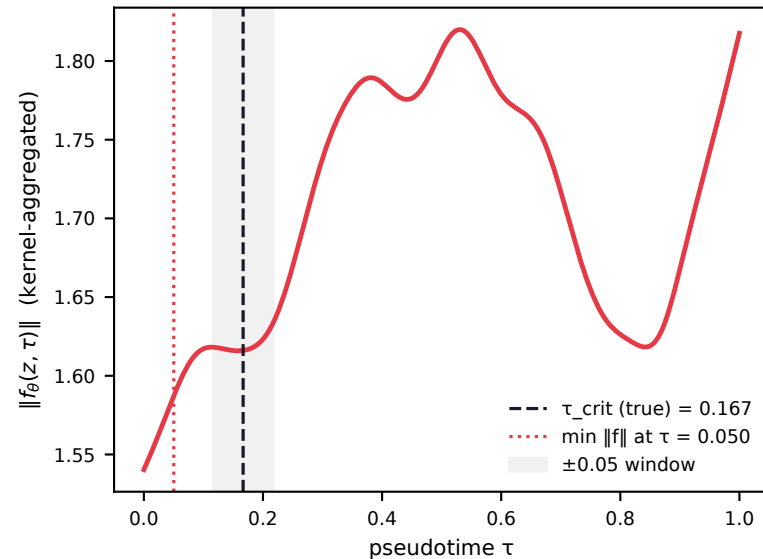


Peak location shifts monotonically $0.60 \rightarrow 0.95$ across $h \in [0.01, 0.15]$; only $h = 0.01$ gives a positive peak eigenvalue on this simplified pipeline (3344-cell MET subset, 1000 epochs). Any timing claim should be checked against a bandwidth sweep on the exact same pipeline.

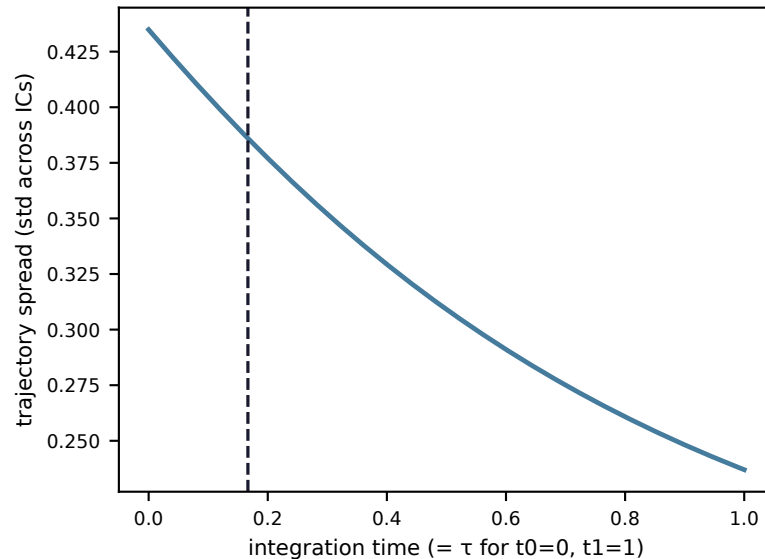
A | Peak τ vs bandwidth**B | Peak $\text{Re}(\lambda_{\text{max}})$ vs bandwidth
(only $h \approx 0.01$ gives a positive peak eigenvalue)**

Test 1 — flow-topology saddle localization on bifurcation_v3 (label-blind) → FAIL

a drift-magnitude collapse [FAIL]



b trajectory spread (raw)



c divergence rate [FAIL]

